Computational imaging

Taras Holotyak

Content of course

The purpose of the course is to introduce into the subject together with providing of a practical skills in the domain of computational imaging.

The course will cover the following topics:

- Computational imaging principles
- Overview of the state of the art of modern computational imaging
- Deep analysis of computational imaging applications

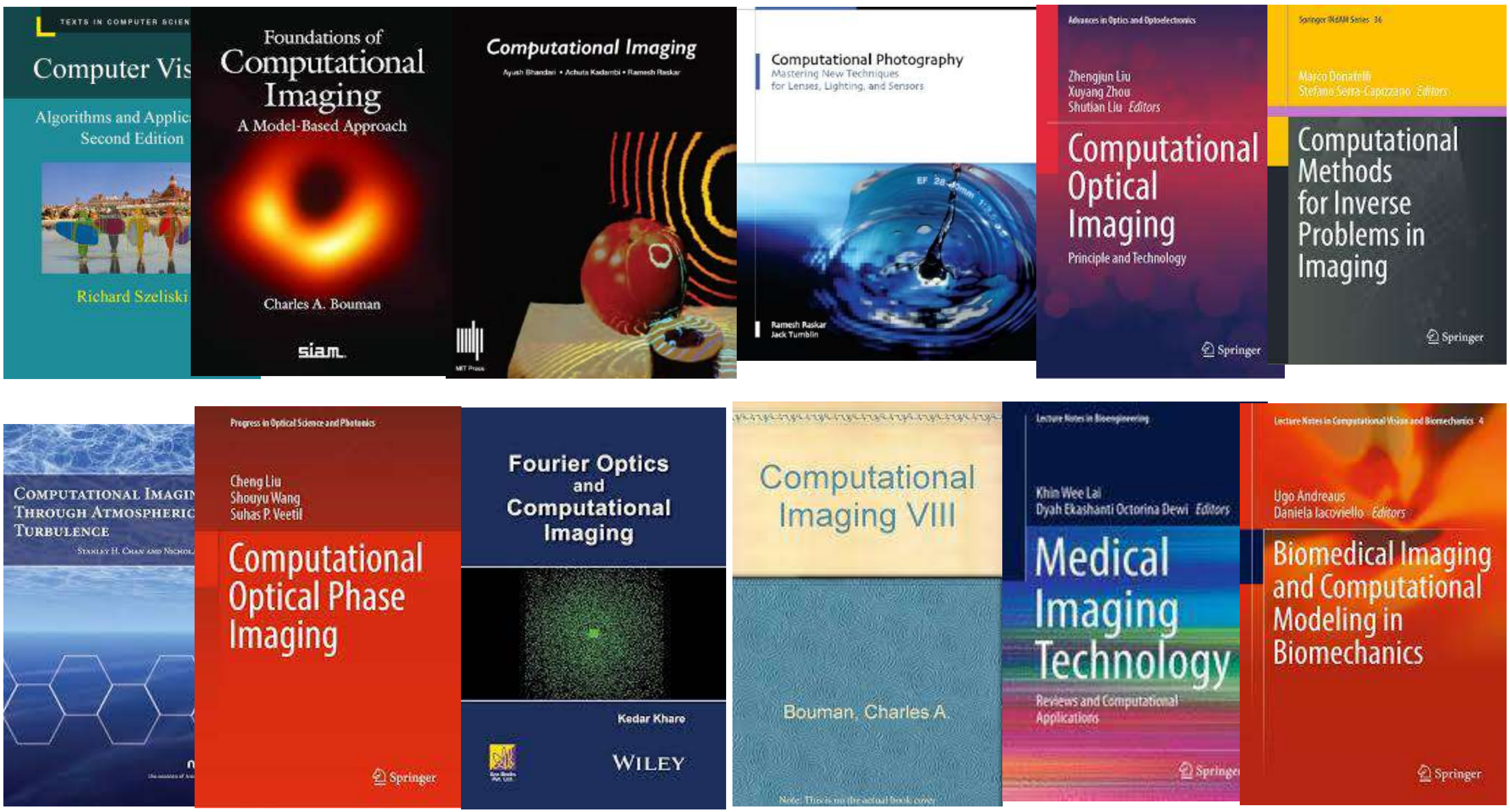
Course grades

Theoretical part (2 CCs/Oral exam): 40%	
TPs: 25%	Mini-project (35%)

- Theoretical part of the course will be evaluated based on 2 CCs or oral exam
- Evaluation of the practical part of the course will combine:
  - Everage grade of the TPs
  - Everage grade of mini-project (grades for report and presentation)

! Grades of all parts of evaluation (theoretical part, TPs, mini-project) have to be positive ≥ 4.0/6/0

Recommended books



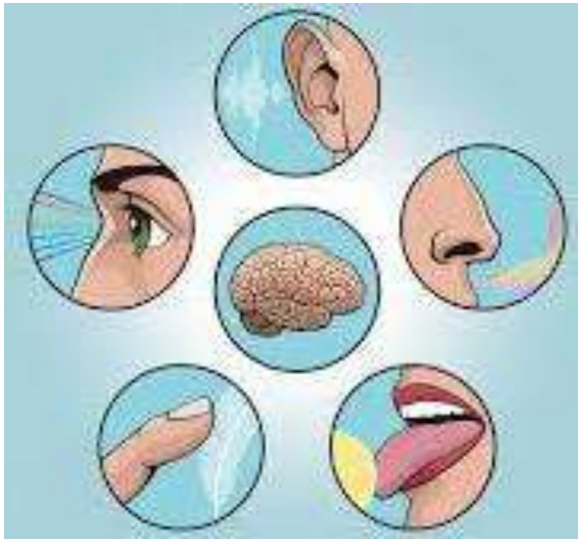
Content

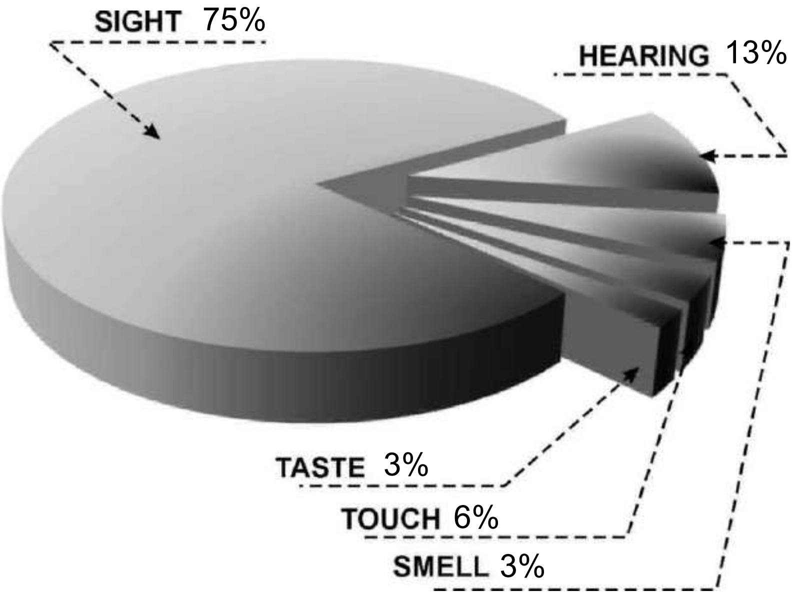
- Introduction to computational imaging
- Motivating examples in computational imaging
- Main topics of computational imaging

Introduction to computational imaging

Computational imaging: why "imaging"?

Human sensing





- About 75% of information is acquired through human visual channel

Introduction to computational imaging

Computational imaging: why “computational”?

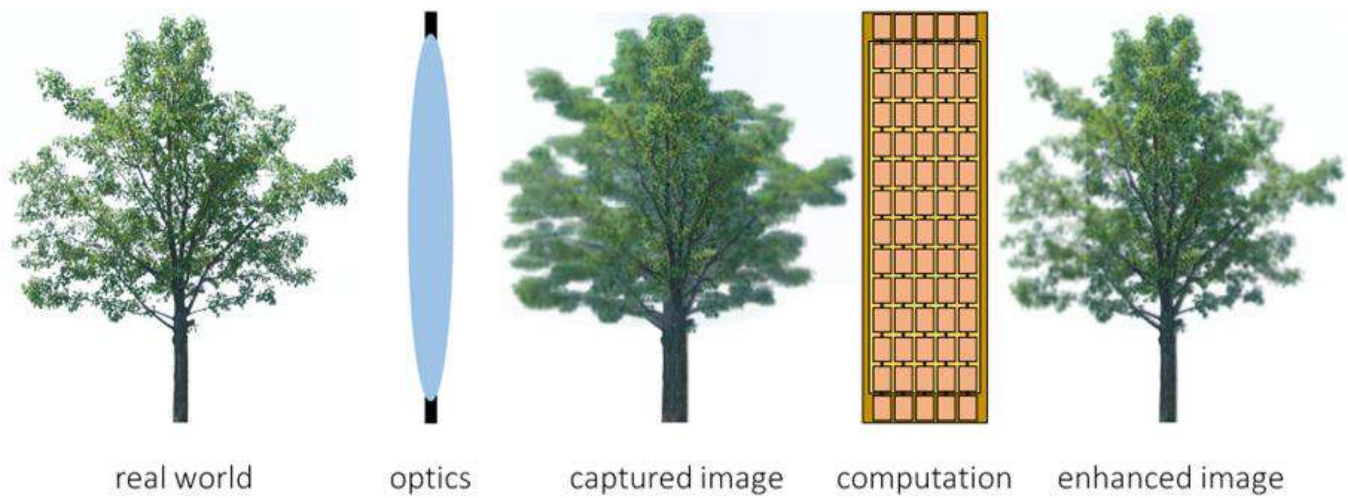
Computational imaging is the process of indirectly forming images from measurements using algorithms that rely on a significant amount of computing. In contrast to traditional imaging, computational imaging systems involve a tight integration of the sensing system and the computation in order to form the images of interest.

Primary goal of **computational imaging** is to reconstruct human-recognizable images from measured data using advanced algorithms while for traditional imaging it is to process already-recognizable images to improve the quality or extract some information.

Introduction to computational imaging

Computational imaging: why "computational"?

Traditional imaging

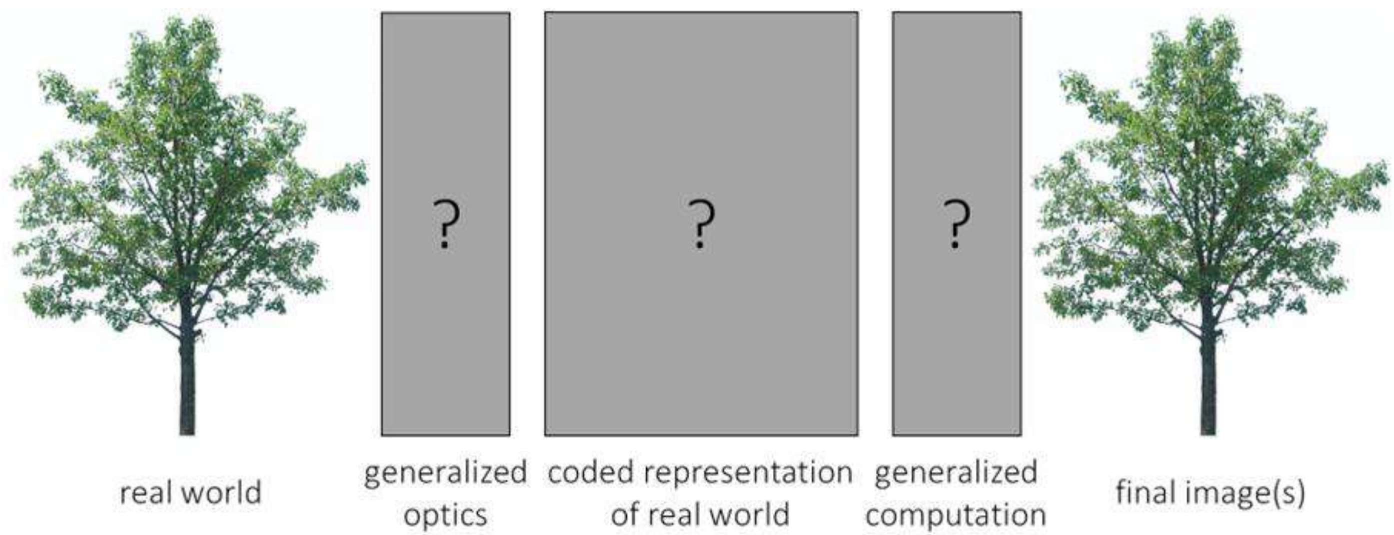


- Optics capture something that is (close to) the final image.
- Computation mostly "enhances" captured image (e.g., deblur).

Introduction to computational imaging

Computational imaging: why "computational"?

Computational imaging

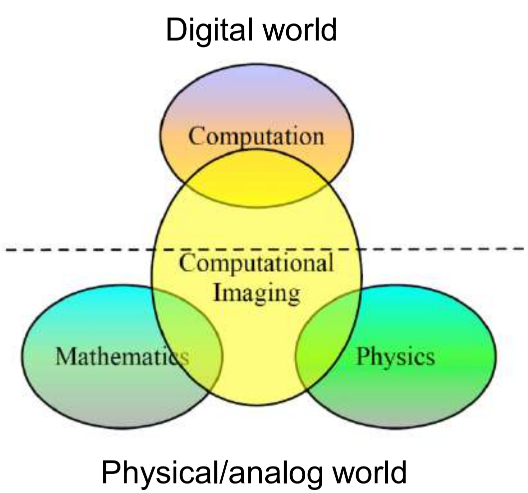


- Generalized optics encode world into intermediate representation.
- Generalized computation decodes representation into multiple images.

Introduction to computational imaging

What is computational imaging?

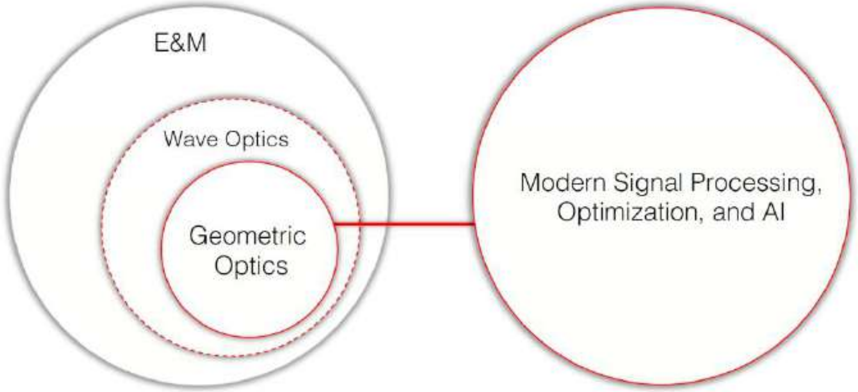
The revolution in sensing, with the emergence of many new imaging techniques, offers the possibility of gaining unprecedented access to the physical world, but this revolution can only be performed through the skillful interplay between the physical and digital worlds. This is the domain of computational imaging which advocates that, to develop effective imaging systems, it will be necessary to go beyond the traditional decoupled imaging pipeline where device physics, image processing and the end-user application are considered separately. Instead, we need to rethink imaging as an integrated sensing and inference model.

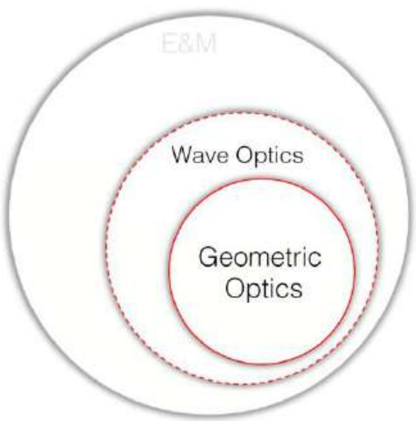


- Access to the physical/analog and digital worlds
- Consideration of imaging as an integrated sensing and inference model

Introduction to computational imaging

What is computational imaging?

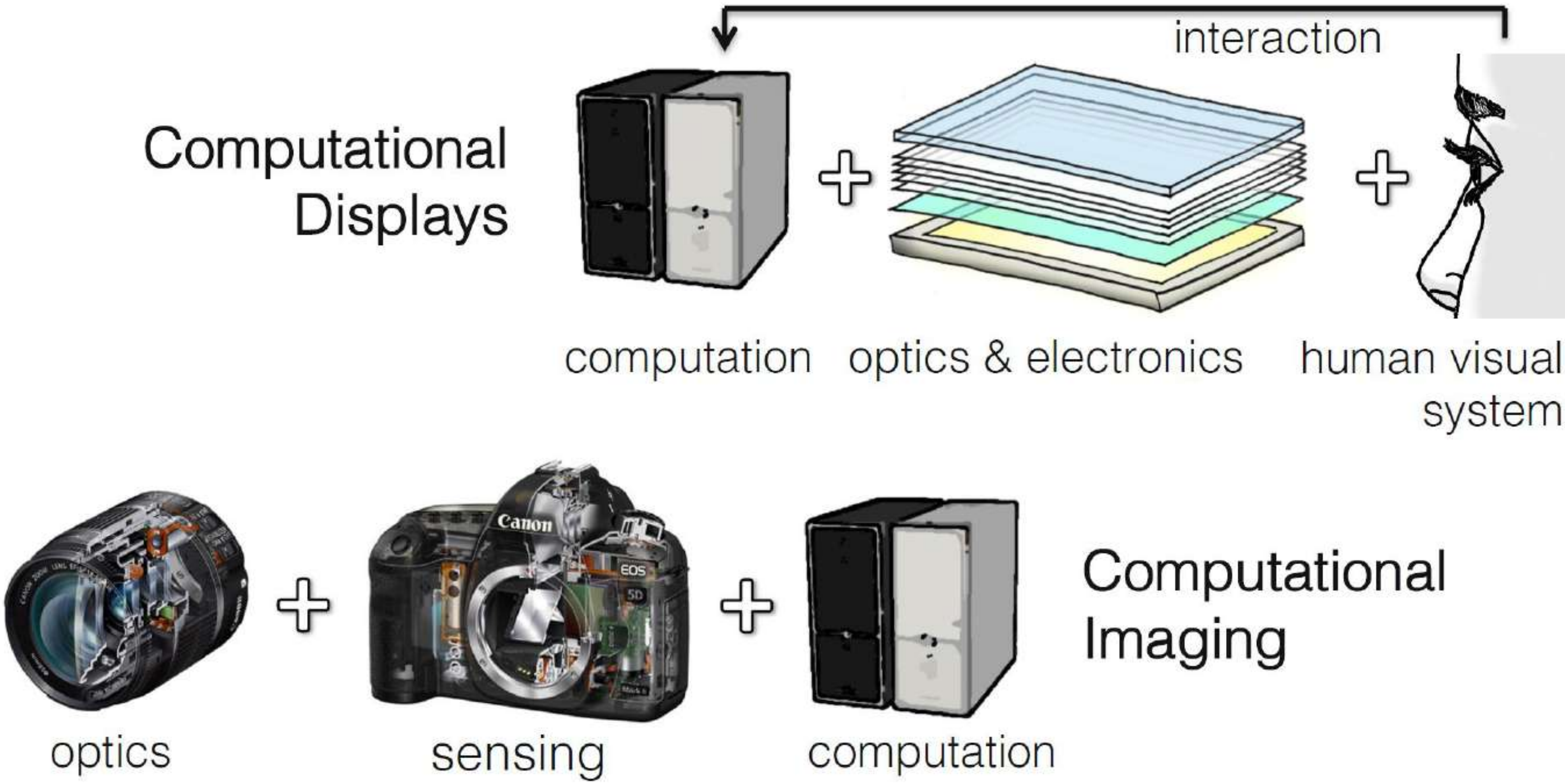




- light as rays
- unit: (spectral) radiance
- properties: wavelength, polarization, direction, …
- only brief introduction & outlook for wave optics

Introduction to computational imaging

What is computational imaging?

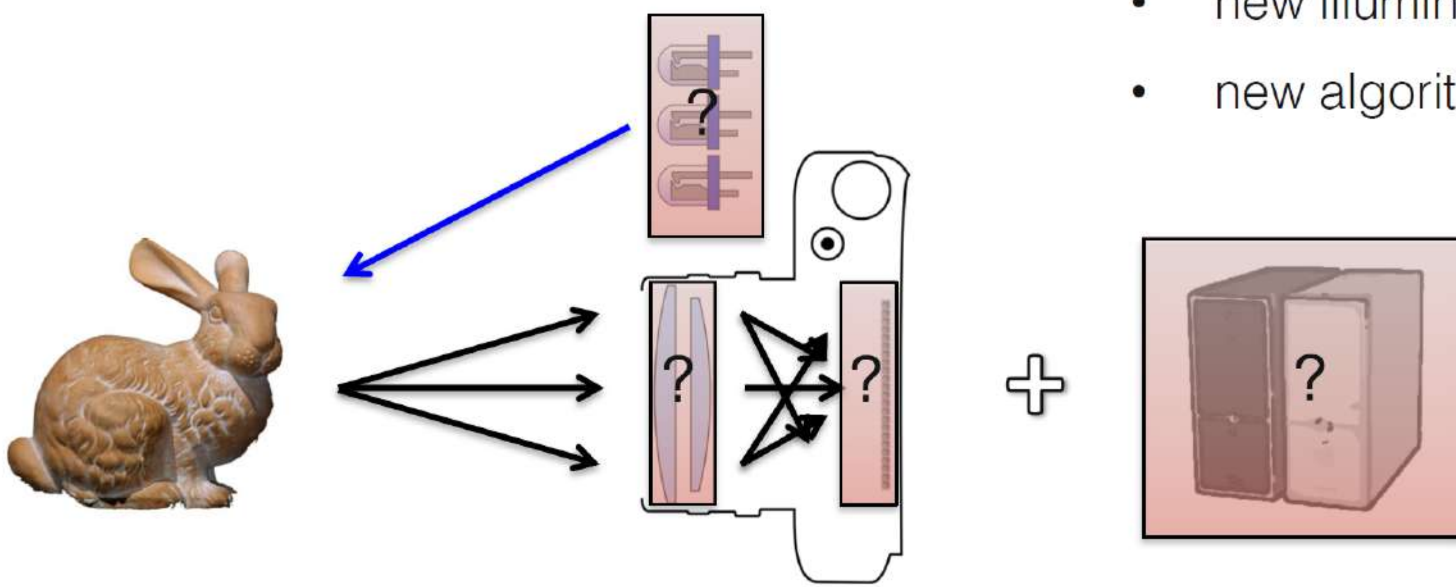


Introduction to computational imaging

What is computational imaging?

1. optically encode scene information
2. computationally recover information

- new optics
- new sensors
- new illumination
- new algorithms

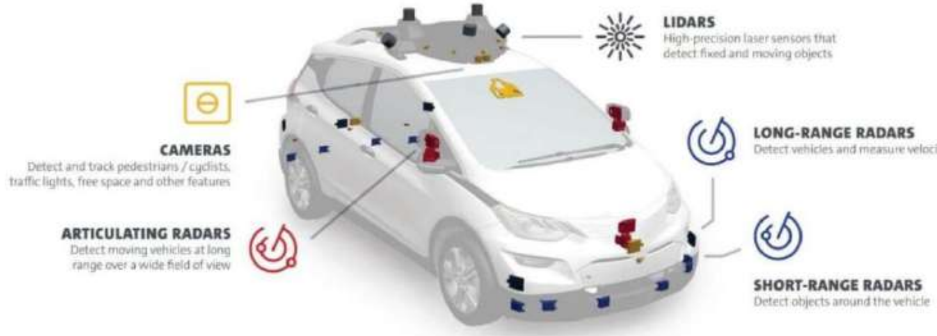


Introduction to computational imaging

Example of biological computational imaging

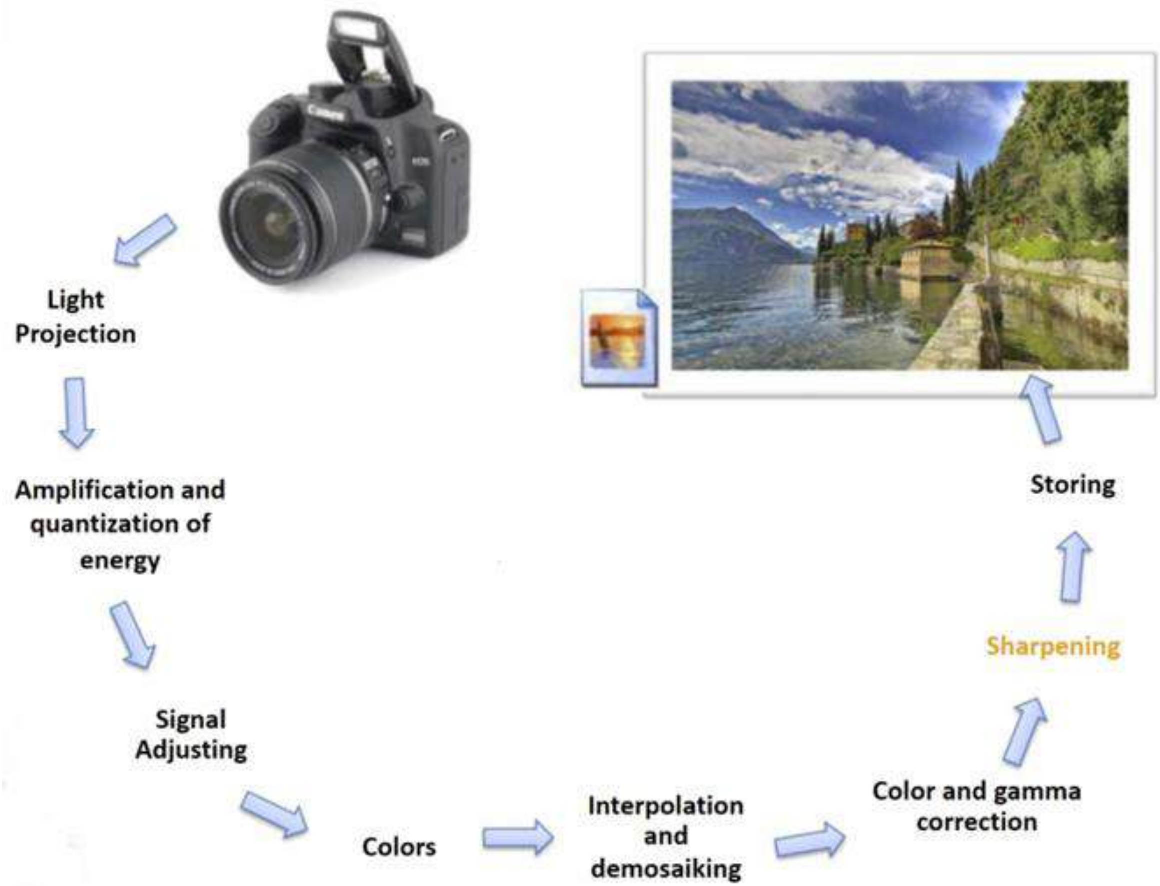
Introduction to computational imaging

Areas of modern computational imaging

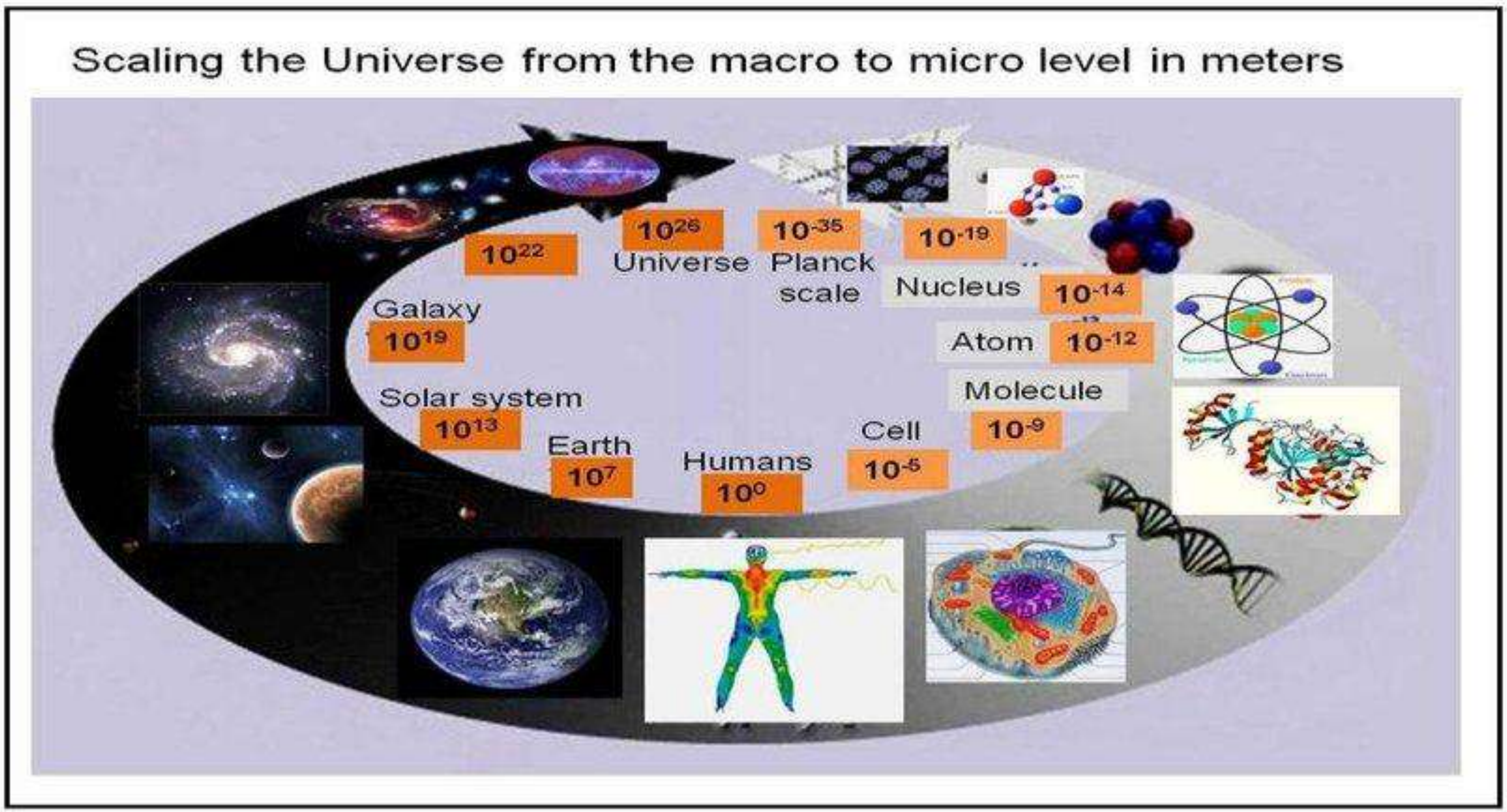


Introduction to computational imaging

Digital photography as example of computational imaging



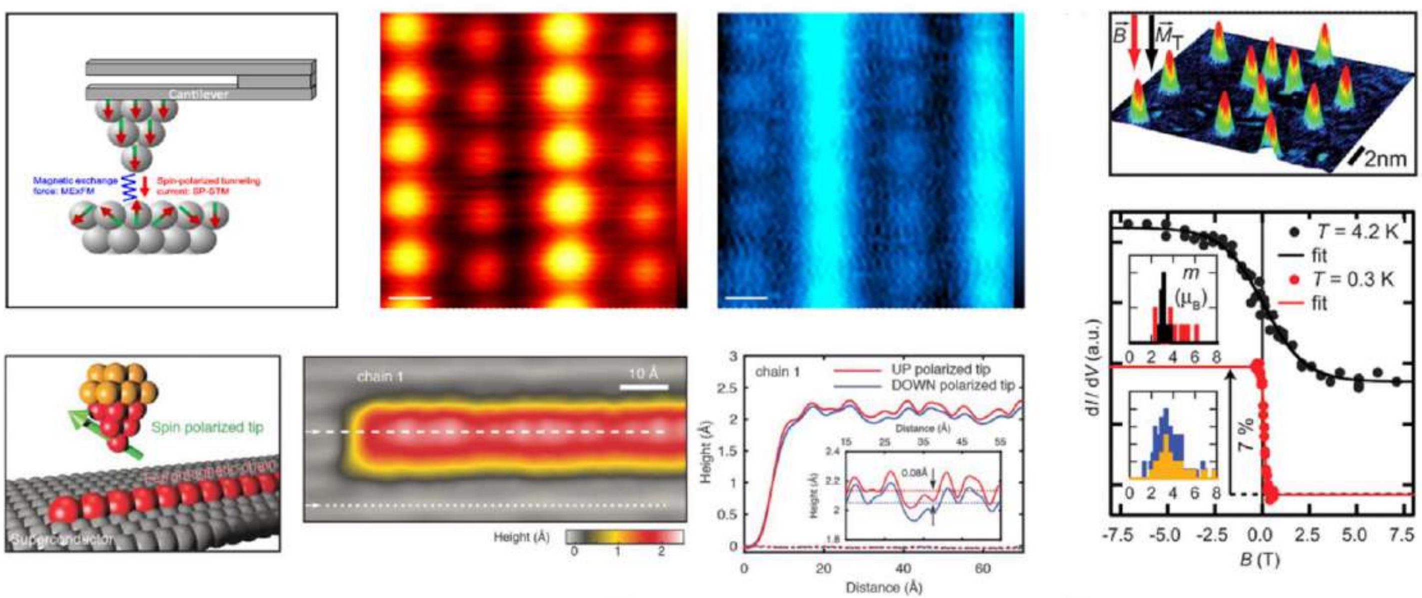
Motivating examples in computational imaging



Motivating examples: nano-scale microscopy

Spin imaging on the atomic scale with STM and non-contact AFM

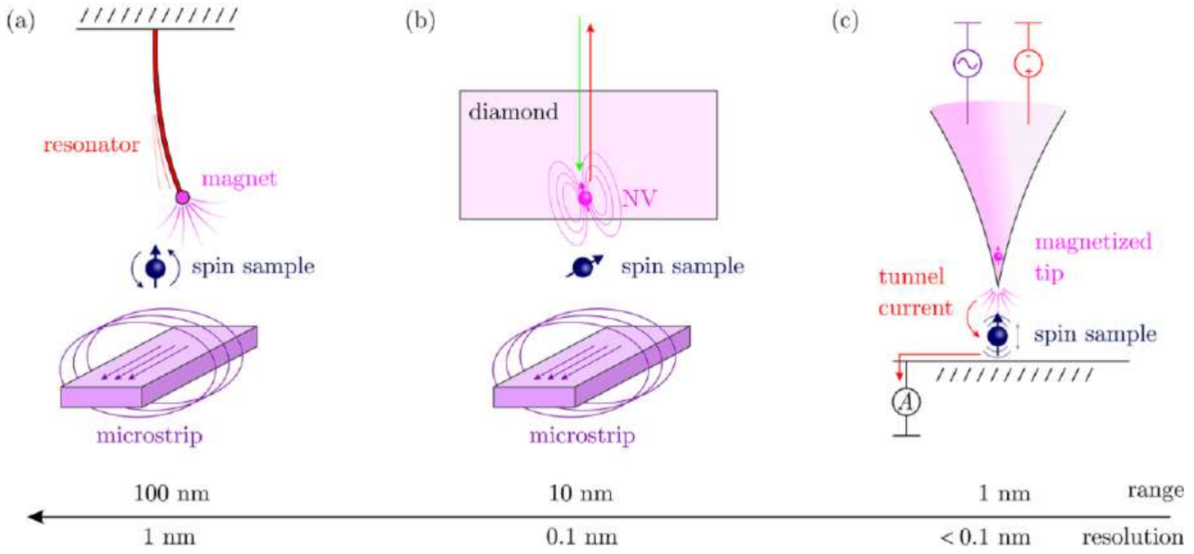
Spin sensing on surfaces with atomic-scale resolution has been achieved using two different, but compatible methods: spin-polarised scanning tunnelling microscopy (SP-STM), and non-contact atom force microscopy (nc-AFM). SP-STM-based spin sensing methods rely on local detection of a spin-component of the tunnelling current, whereas nc-AFM-based spin sensing methods rely on the detection of local forces related to magnetic exchange.



Motivating examples: nano-scale microscopy

Nanoscale magnetic resonance imaging

Magnetic resonance force microscopy (MRFM) relies on mechanical force sensors to detect the weak magnetic signal of single nuclear spins.

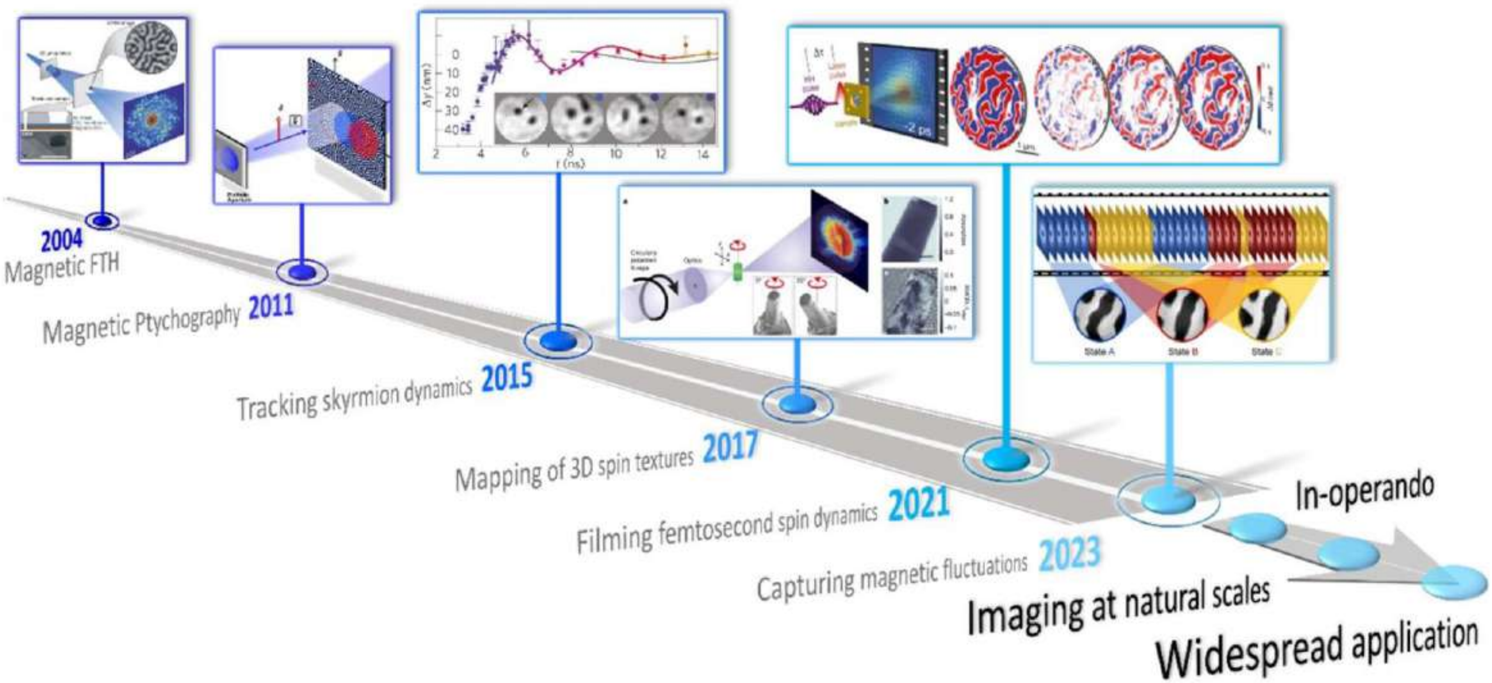


(a) MRFM uses a nanomechanical resonator together with a nanoscale magnet to generate a force on the spins in the sample. A microstrip serves as an antenna that generates radio-frequency pulses for NMR spin inversion
(b) Nitrogen vacancy (NV)-based NanoMRI: an electronic defect spin situated within nanometres from the diamond surface acts as the magnetic sensor
(c) In Electron Spin Resonance-Scanning Tunneling Microscopy (ESR-STM), a tunnel current flowing between the sharp scanning tip and the conducting substrate serves as the signal

Motivating examples: nano-scale microscopy

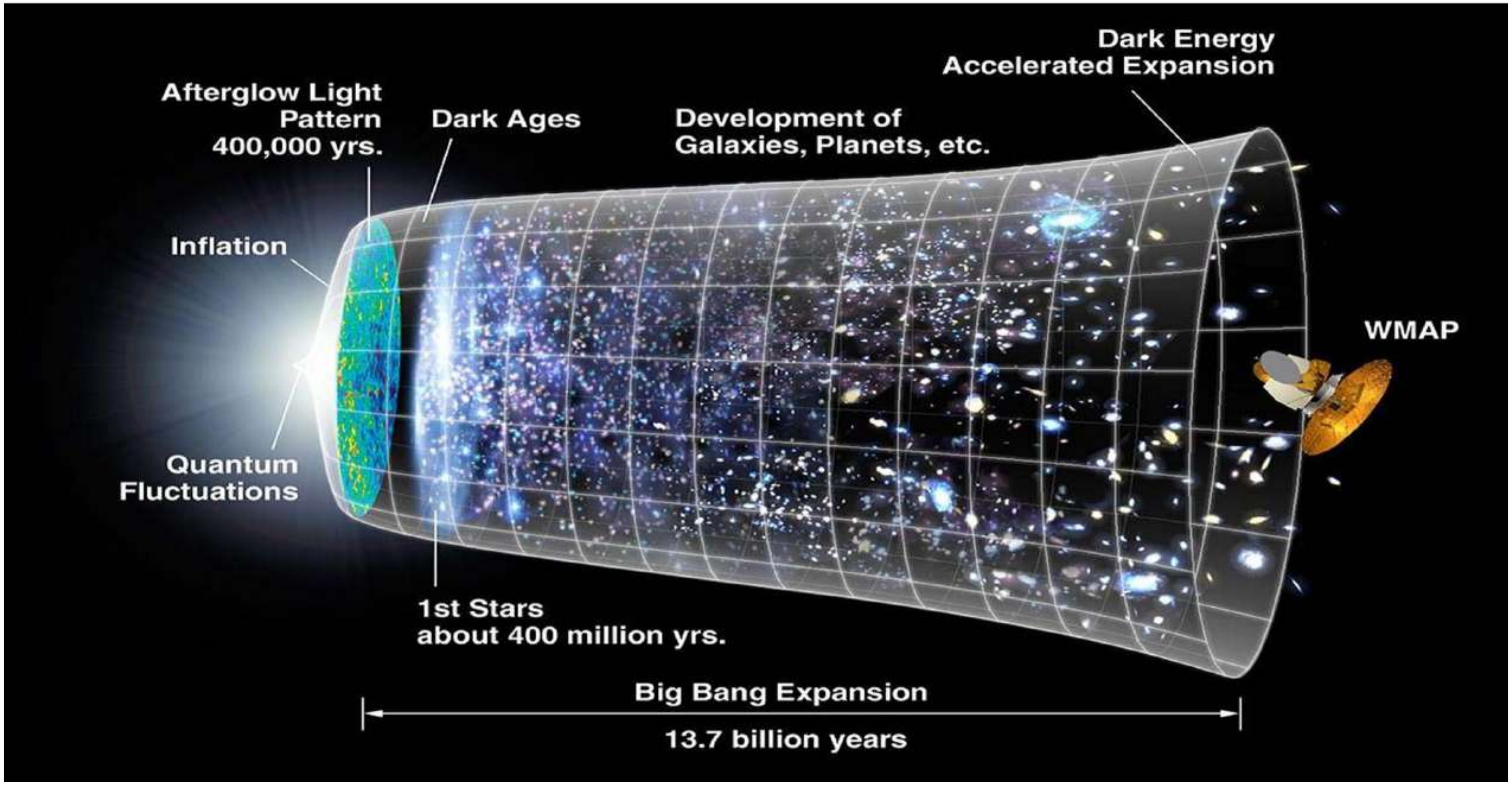
Coherent x-ray imaging of magnetic systems

Coherent x-ray magnetic imaging is a lensless approach to magnetic microscopy. Highlights of this approach are the simultaneous access to the element-specific phase and absorption contrasts and the promise of wavelength-limited spatial resolution (5nm in 2D and 10nm in 3D).



Motivating examples: astro applications

Key outstanding questions: understanding of the Universe



Motivating examples: astro applications

Motivating examples: astro applications

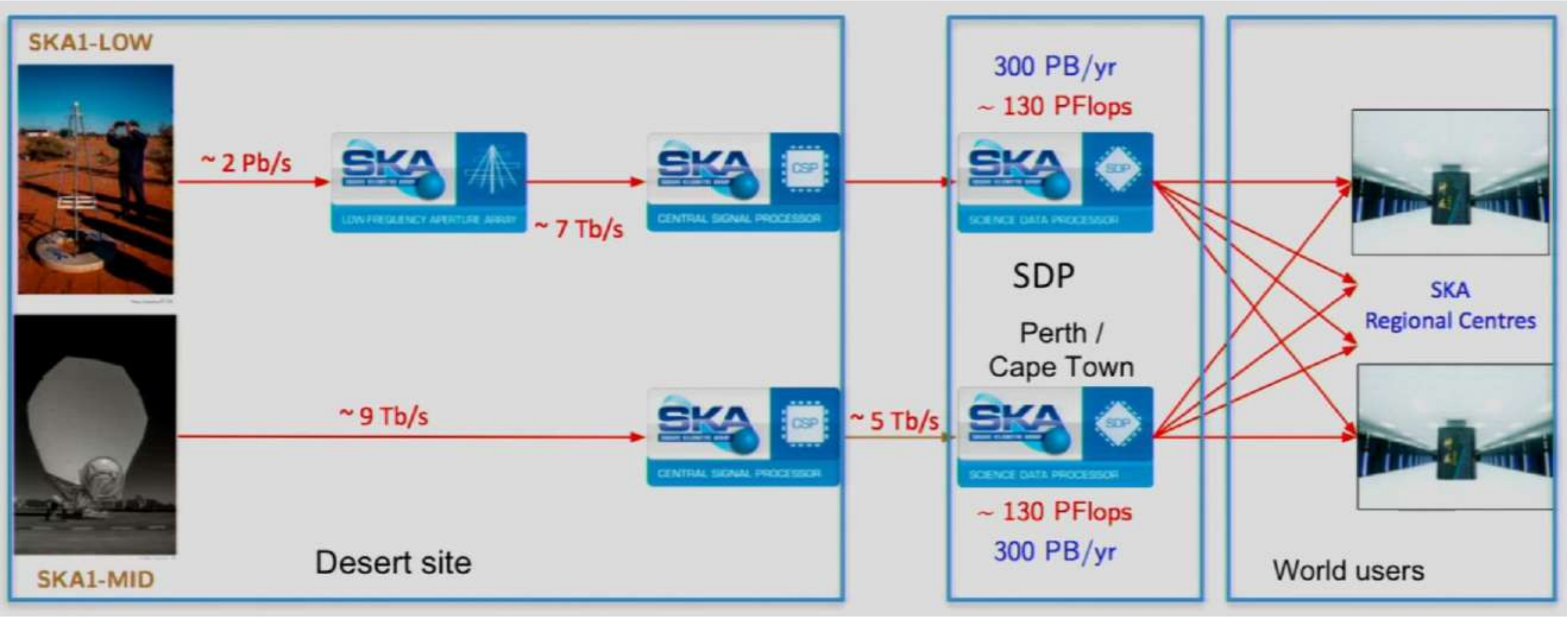
Information acquisition sensors

SKA1-Low: 131,072 low-freq antennas
(512 stations each with 256 dipoles)
50 – 350 MHz
65 km baselines (11″ @ 110 MHz)
Murchison, **Western Australia**

SKA1-Mid: 197 dishes
(133 x 15m + 64 x 13.5m dishes)
0.35 – 15.4 GHz
MeerKAT
150 km baselines
(0.22″ @ 1.7 GHz; 34 mas @ 15 GHz)
Karoo, **South Africa**

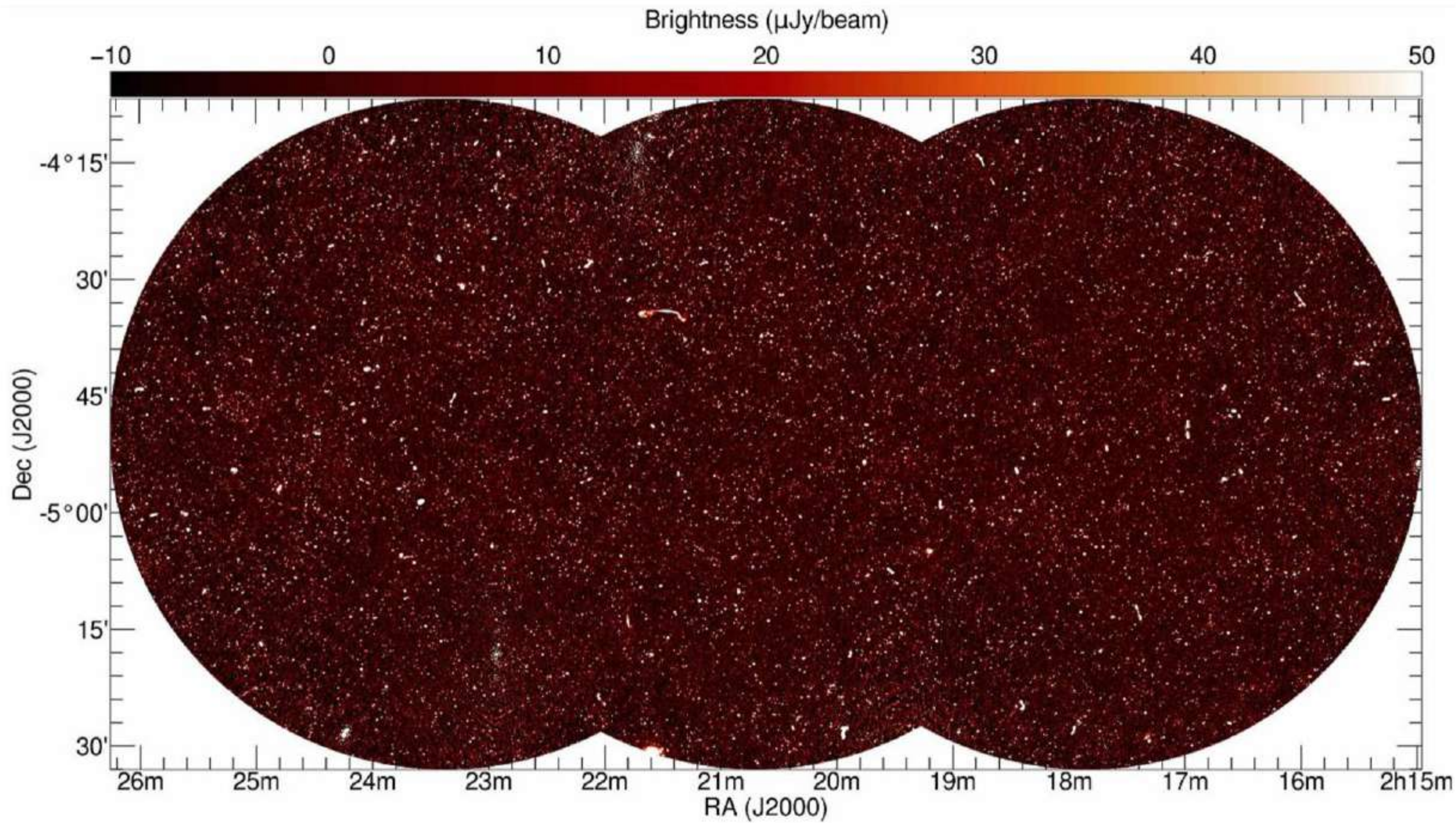
Motivating examples: astro applications

Amount of the generating data



Motivating examples: astro applications

Expected results



Motivating examples: medical imaging

Medical imaging classification

Method	information carrier	note
Ultrasound imaging	high-frequency pressure waves	backscattering / reflection imaging
X-ray imaging (projection radiography)	ionizing radiation	transmission imaging
Computed tomography: CT, C-arm		
Nuclear imaging: SPECT, PET		emission imaging
Magnetic resonance imaging (MRI)	nuclear spin precession	magnetic resonance imaging

Processing steps:

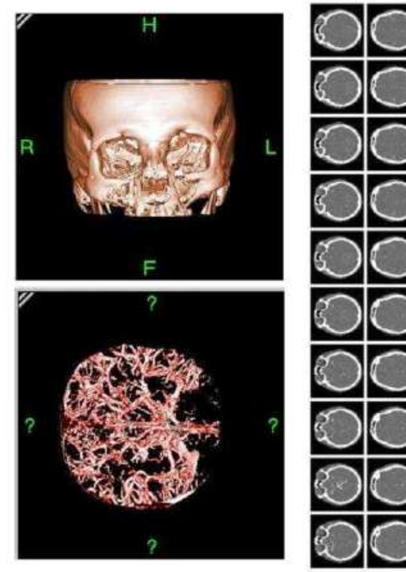
1) Acquisition of multiple 1D and 2D signals

2) Reconstruction of 3D object

Motivating examples: medical imaging

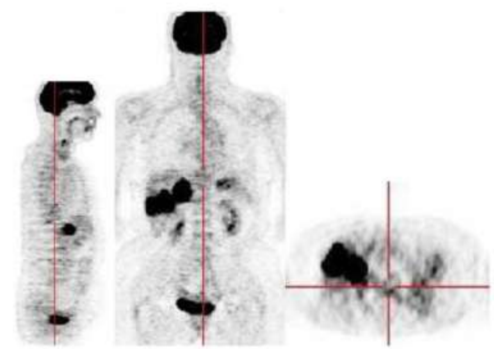
3D ultrasound

Computed Tomography (CT)



Projected X-ray transmission

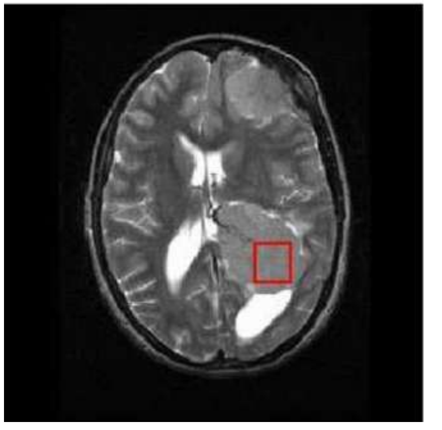
PET



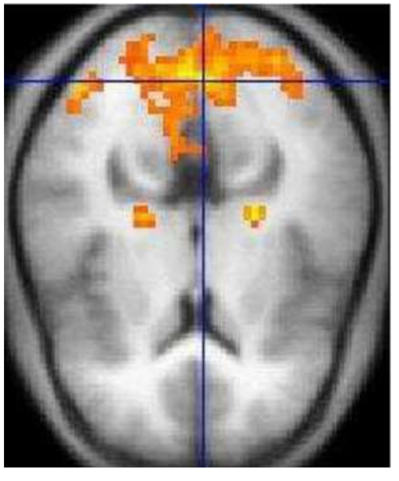
Motivating examples: medical imaging

MR angiography (MRA)

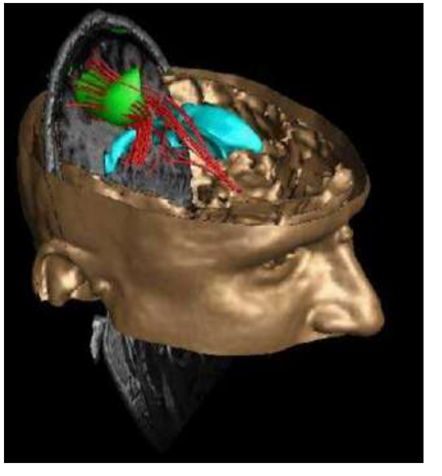
Magnetic resonance spectroscopy (MRS)



Functional MRI (fMRI)



Current density imaging (CDI)



Motivating examples: computational photography

Imaging by mobile devices

Motivating examples: computational photography

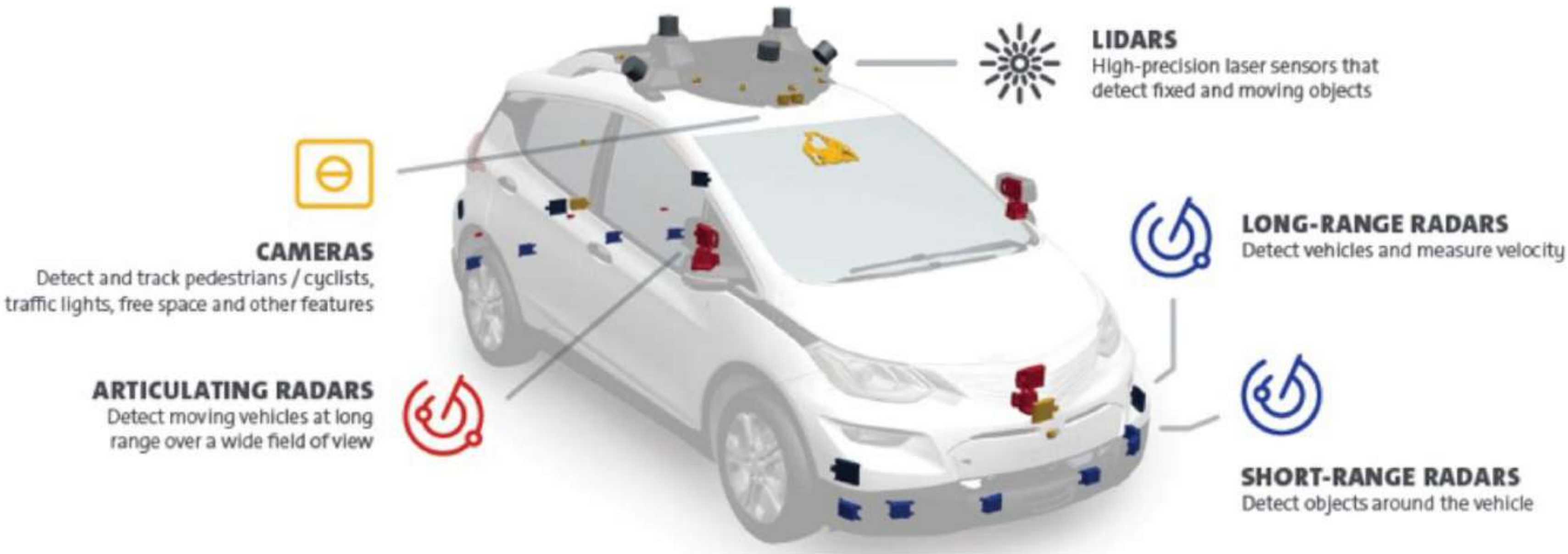
Digital photography

Motivating examples: computational photography

Multi-sensor / hyperspectral imaging

Motivating examples: car industry

Self-driving cars



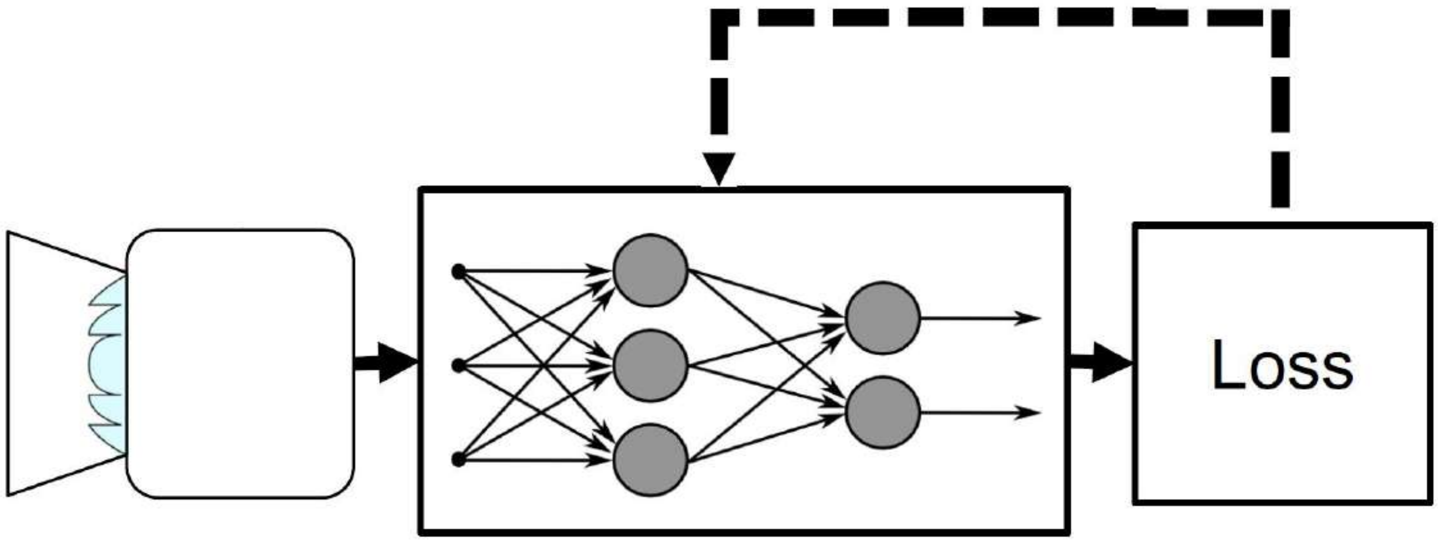
Motivating examples in computational imaging

3D imaging for autonomous vehicles

LIDAR (light detection and ranging)
Velodyne VLS-128

Motivating examples: AI based solutions

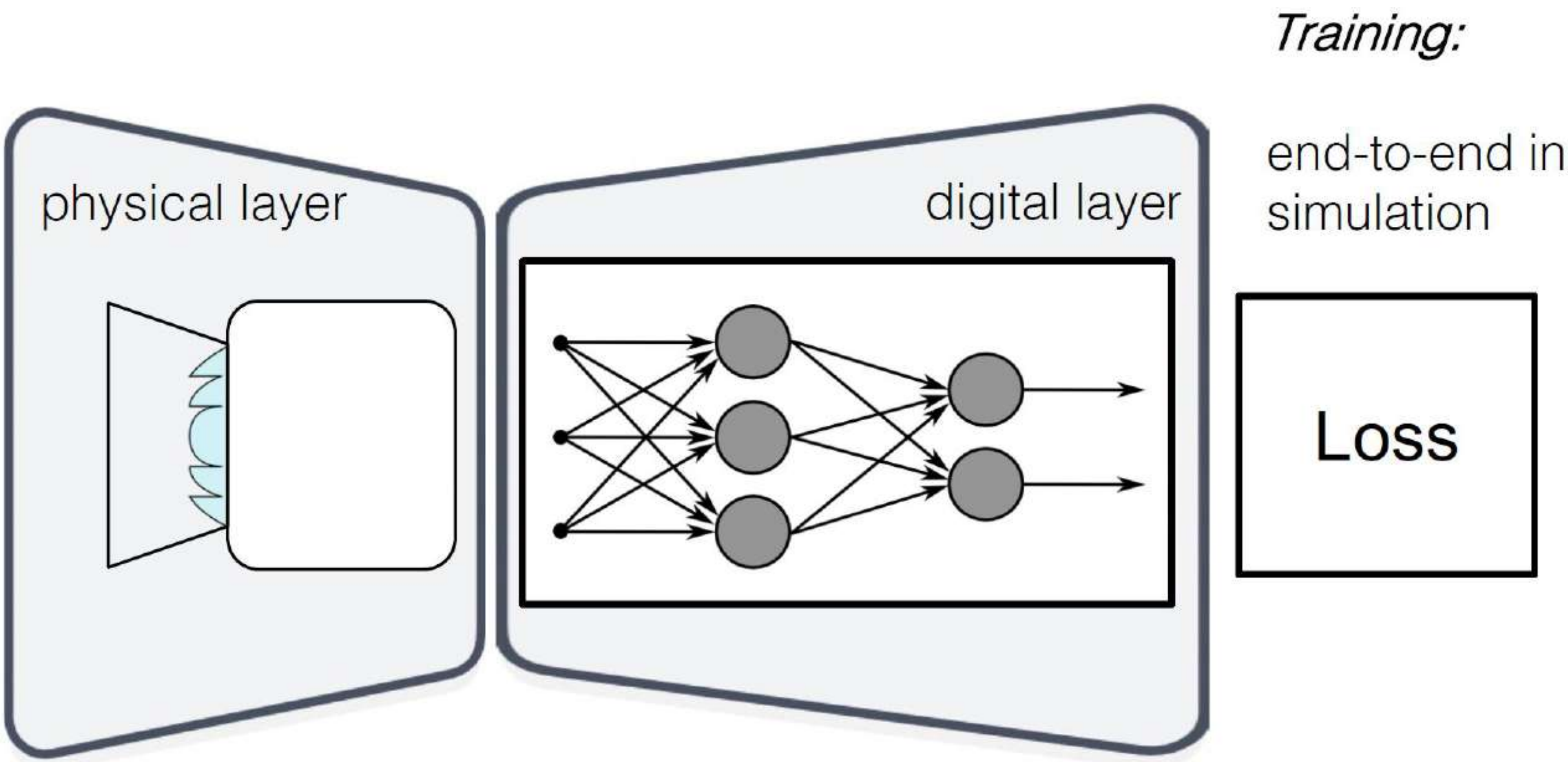
Deep optics



Jointly optimize optics and image processing end-to-end

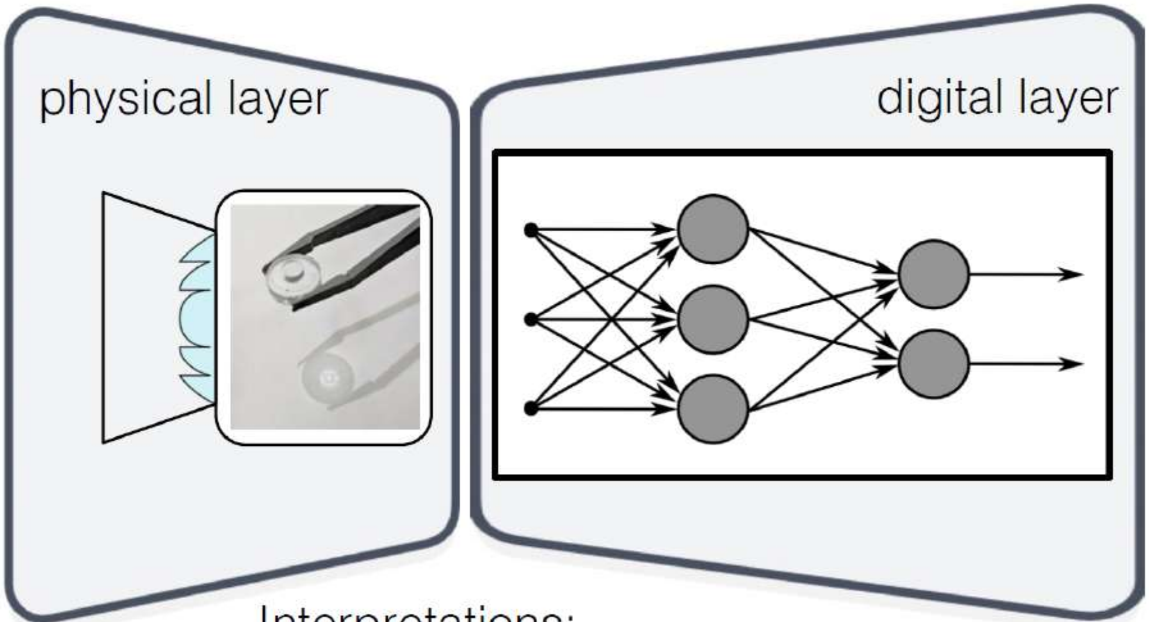
Motivating examples: AI based solutions

Deep optics



Motivating examples: AI based solutions

Deep optics



Inference:

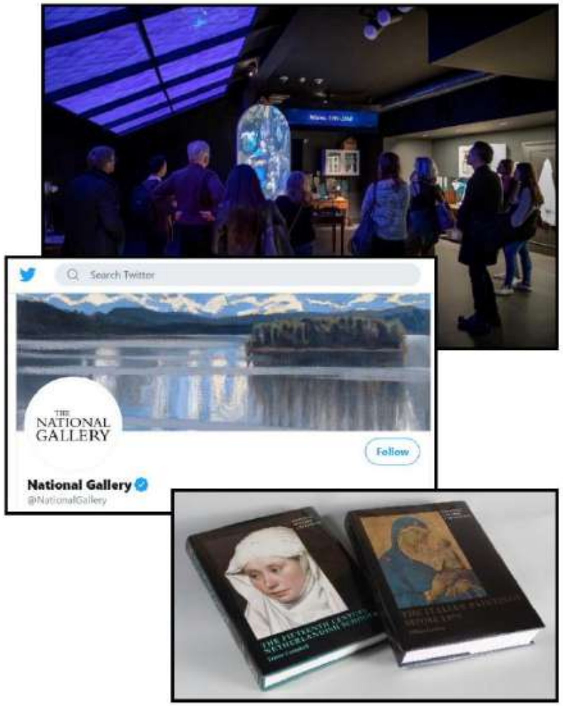
fabricate lens or other physical components, run network

Interpretations:

- Optical encoder, electronic decoder system
- Hybrid optical-electronic neural network

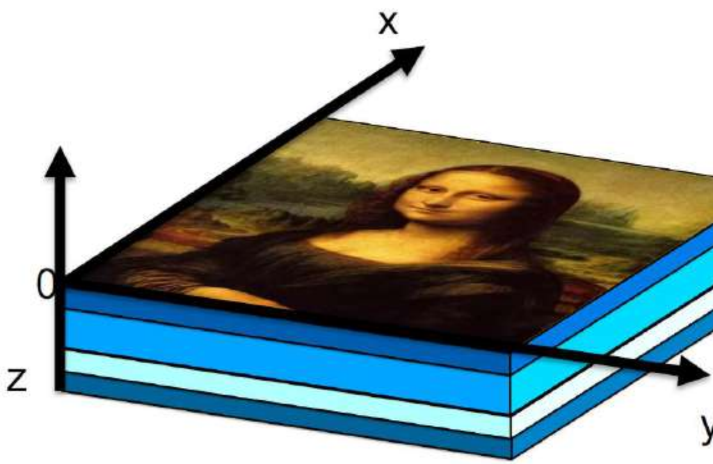
Motivating examples: art analysis

Examination of old painting

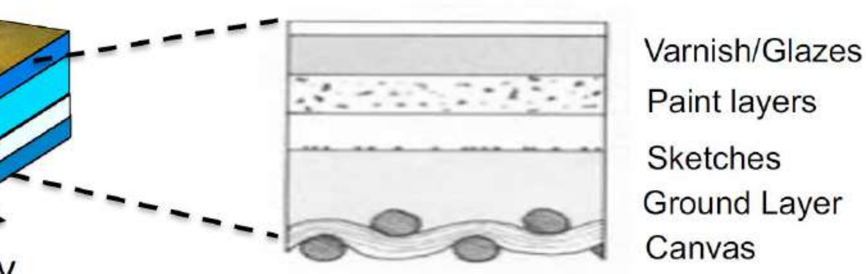


Motivating examples: art analysis

Examination of old painting (structure of painting)



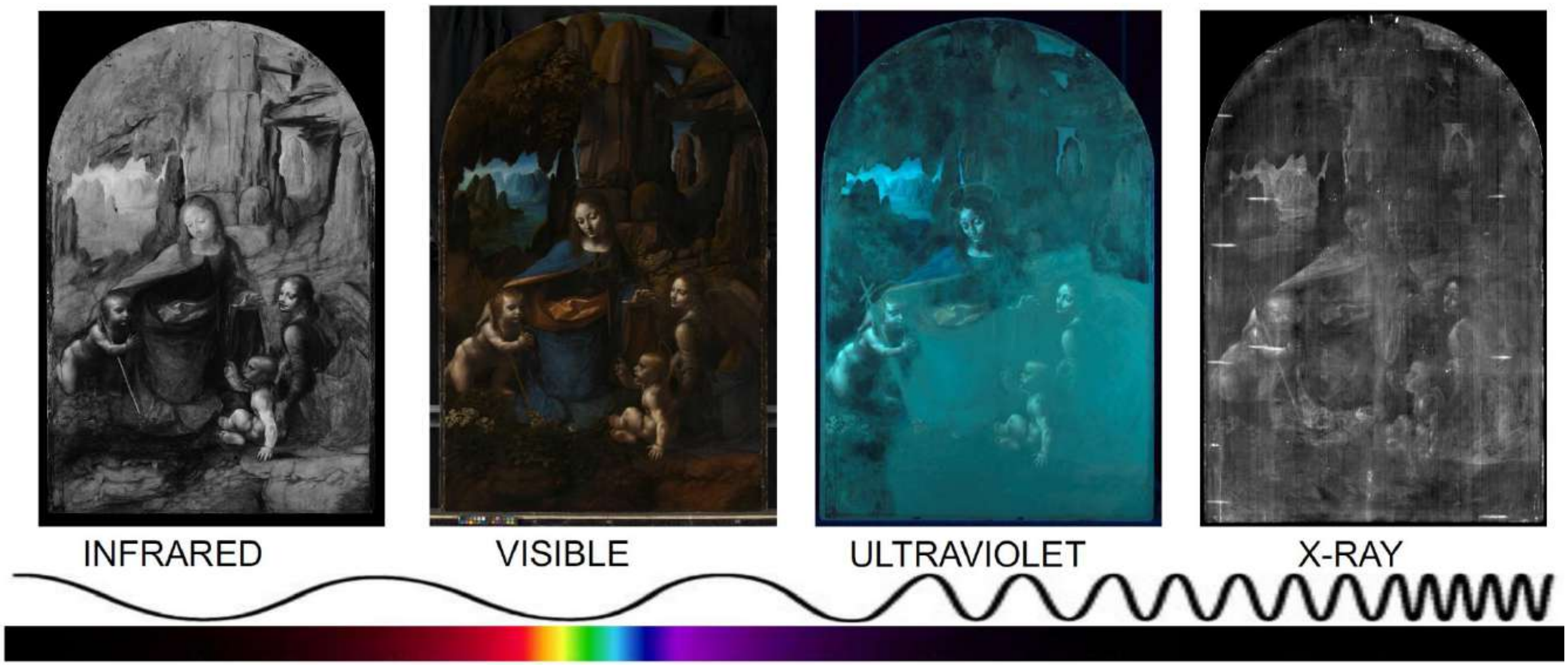
Idealised painting stratigraphy



Realistic painting stratigraphy

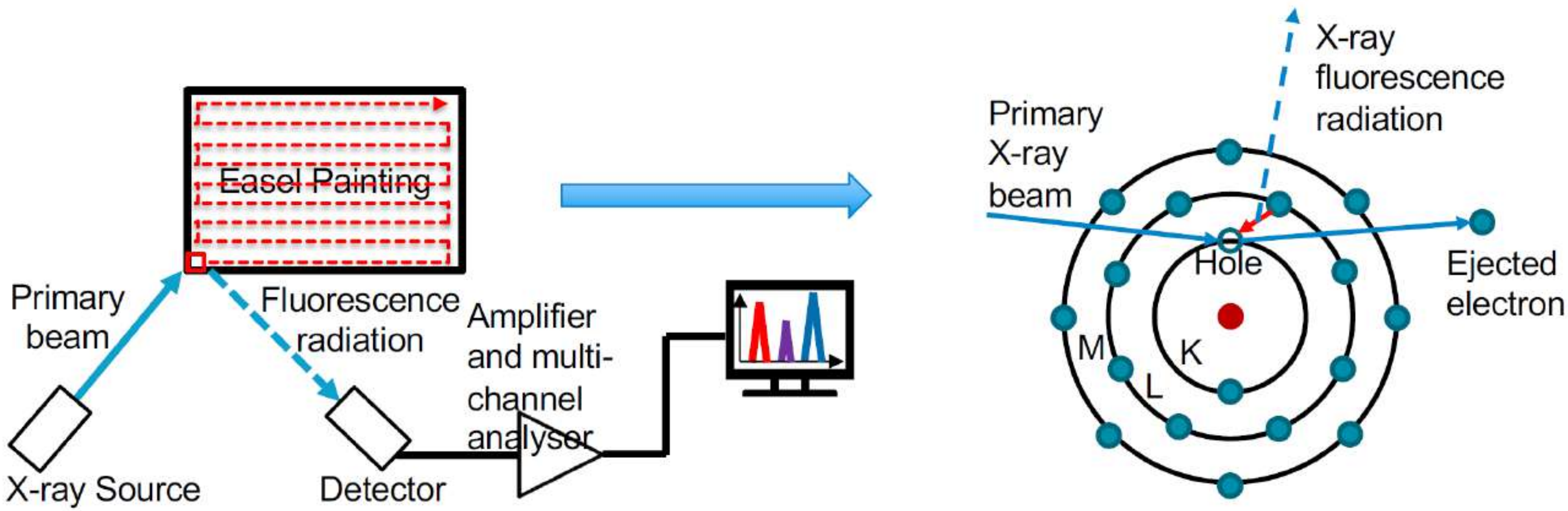
Motivating examples: art analysis

Examination of old painting (traditional non-invasive imaging)



Motivating examples: art analysis

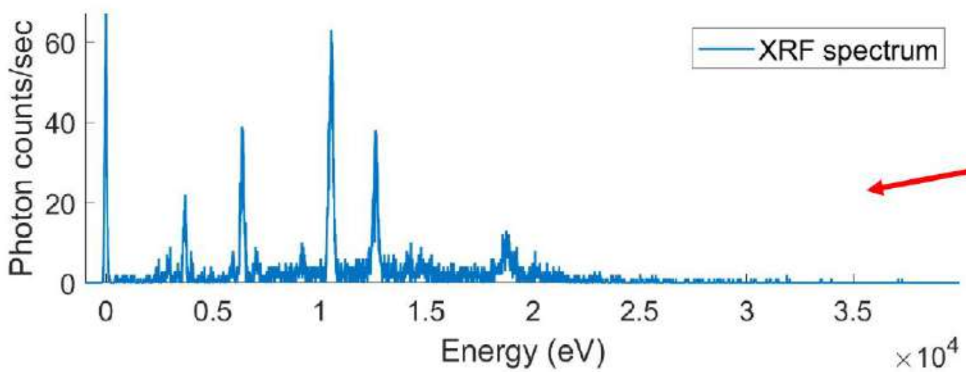
Examination of old painting (macro X-ray fluorescence (MA-XRF))

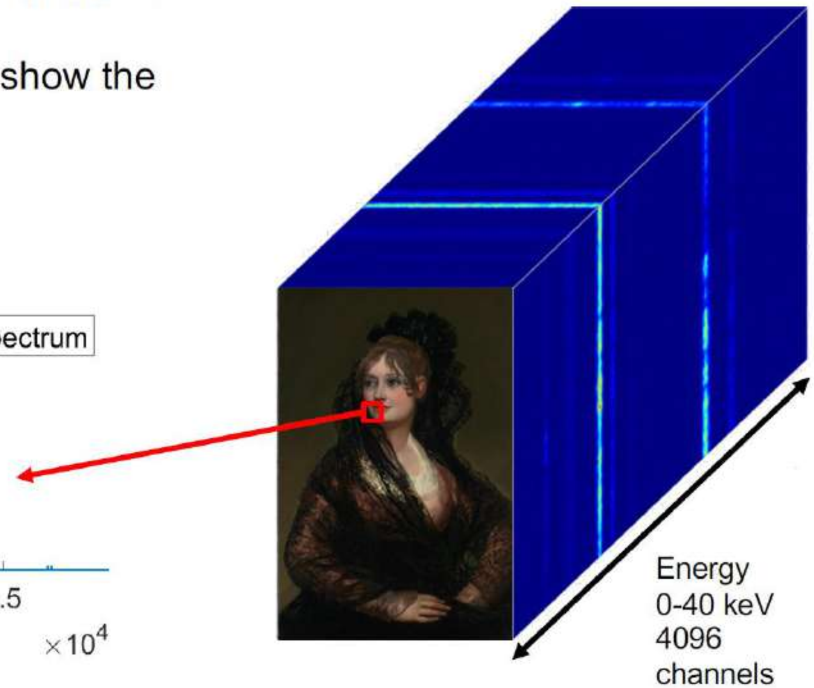


Motivating examples: art analysis

Examination of old painting (MA-XRF datacube and spectrum)

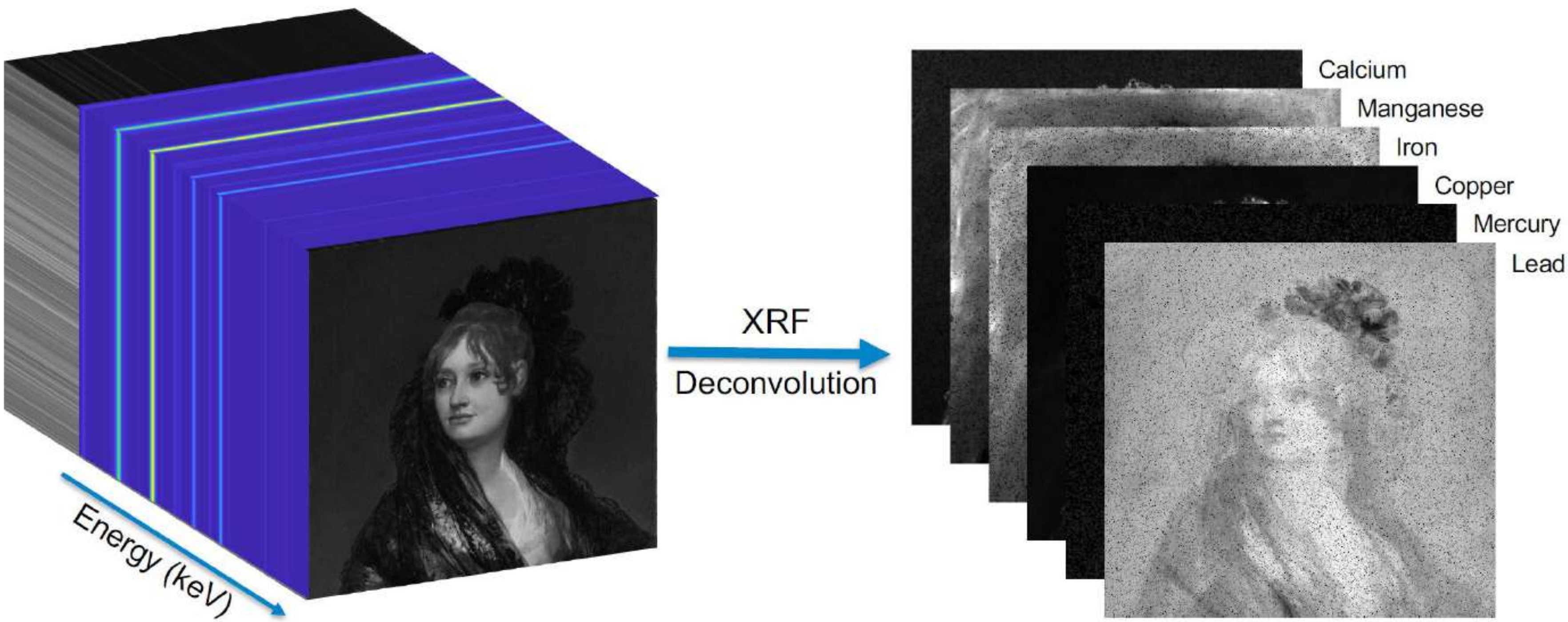
- Macro X-ray provides volumetric data and the locations of the pulses in the energy direction are related to the chemical elements present in the painting.
- This potentially allows us to create maps that show the distribution of different chemical elements





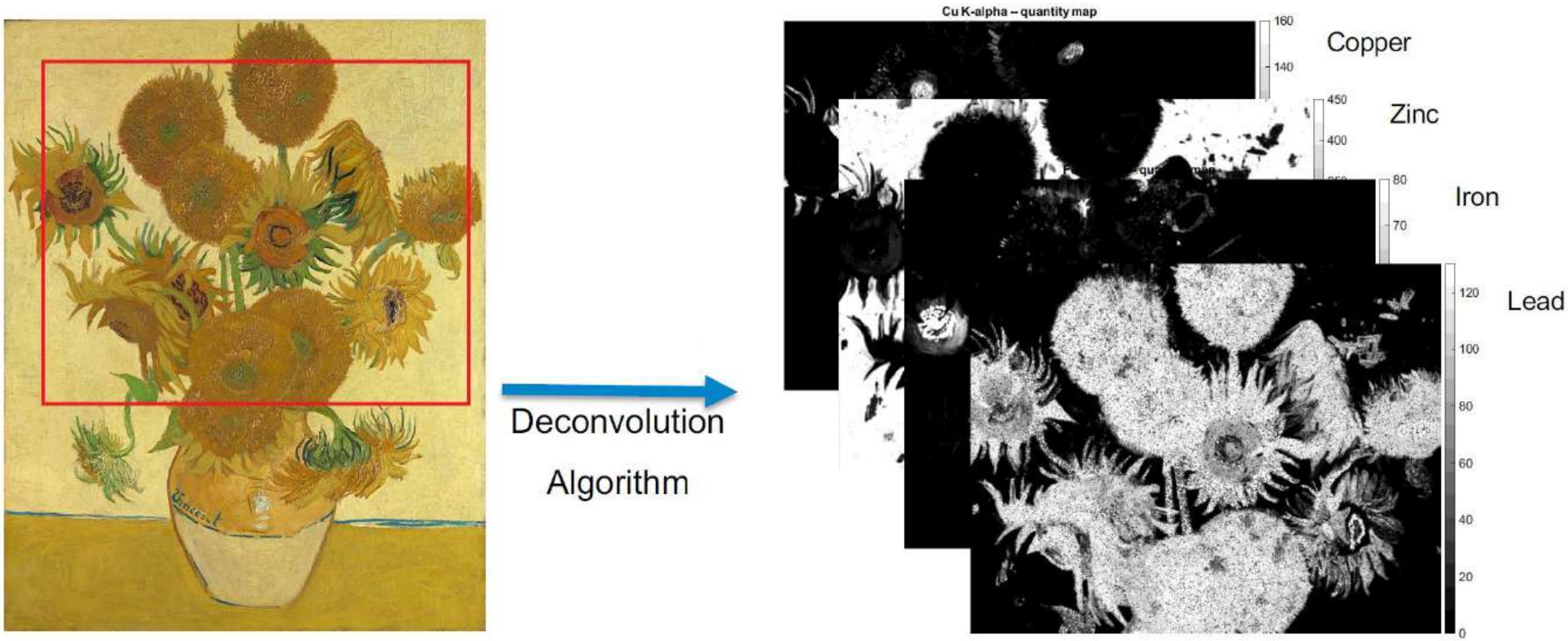
Motivating examples: art analysis

Examination of old painting (extraction of elemental maps)



Motivating examples: art analysis

Examination of old painting (extraction of elemental maps)



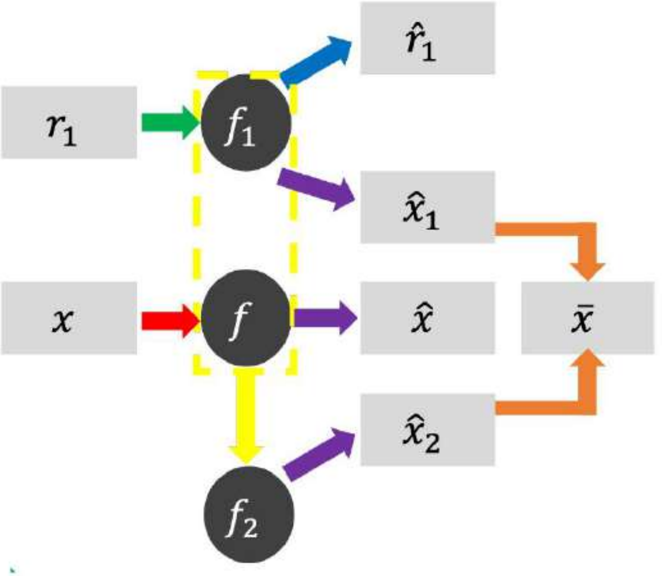
Motivating examples: art analysis

ML based extraction of painting underneath

(a)

(b)

Francisco de Goya, Dona Isabel de Porcel (NG1473), before 1805. Oil on canvas. (a). RGB image. (b). X-ray image.



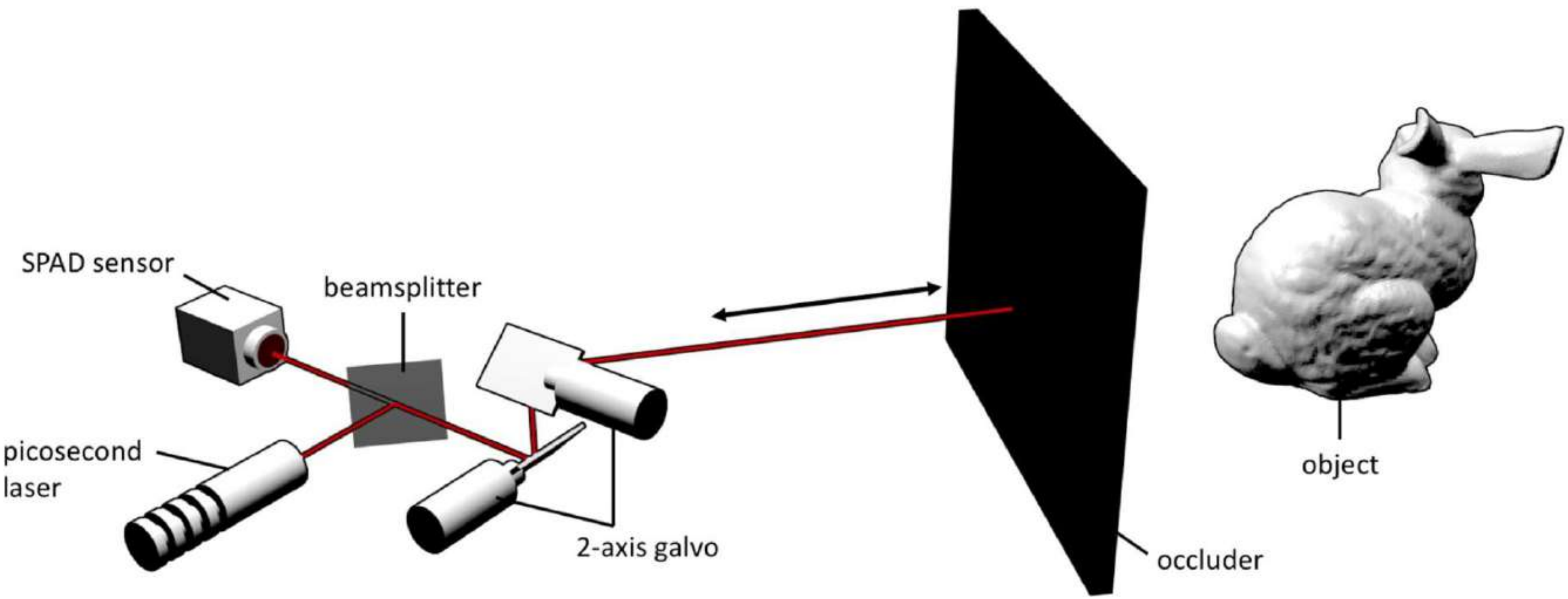
Motivating examples: art analysis

ML based extraction of painting underneath

Separation Results

Motivating examples: non-trivial applications

Non-line-of-sight (NLOS) imaging



Motivating examples: non-trivial applications

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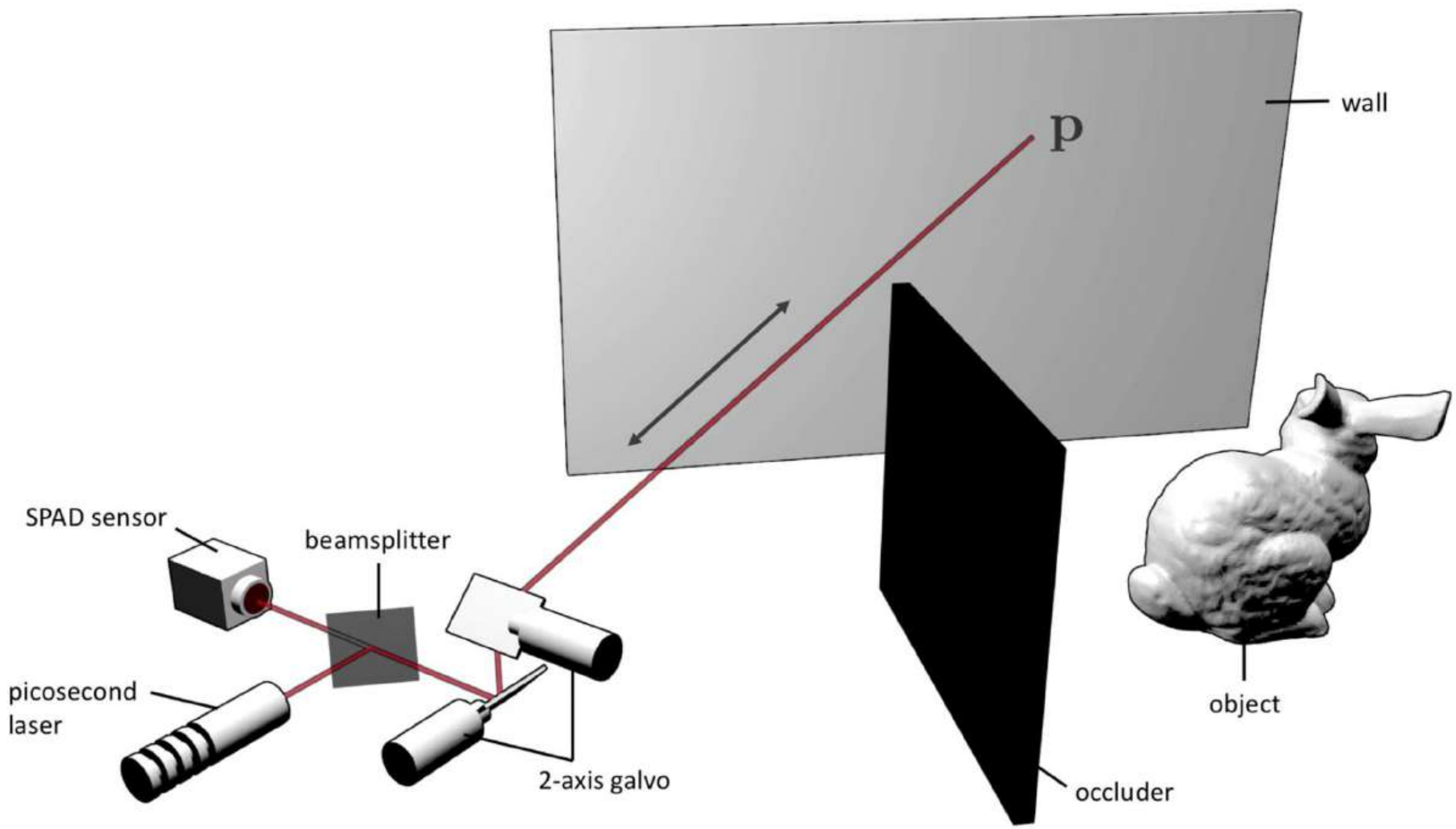

Motivating examples: non-trivial applications

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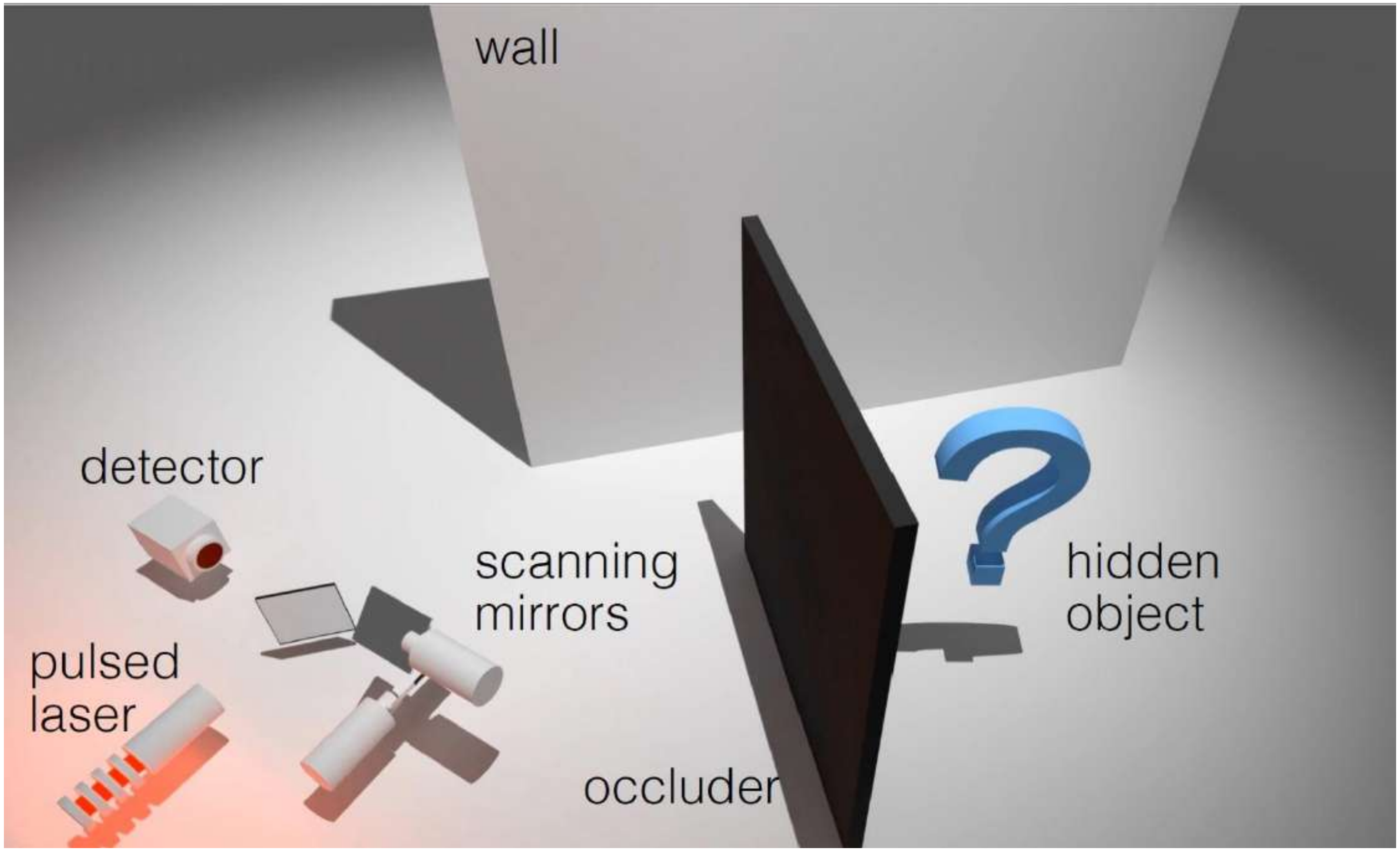

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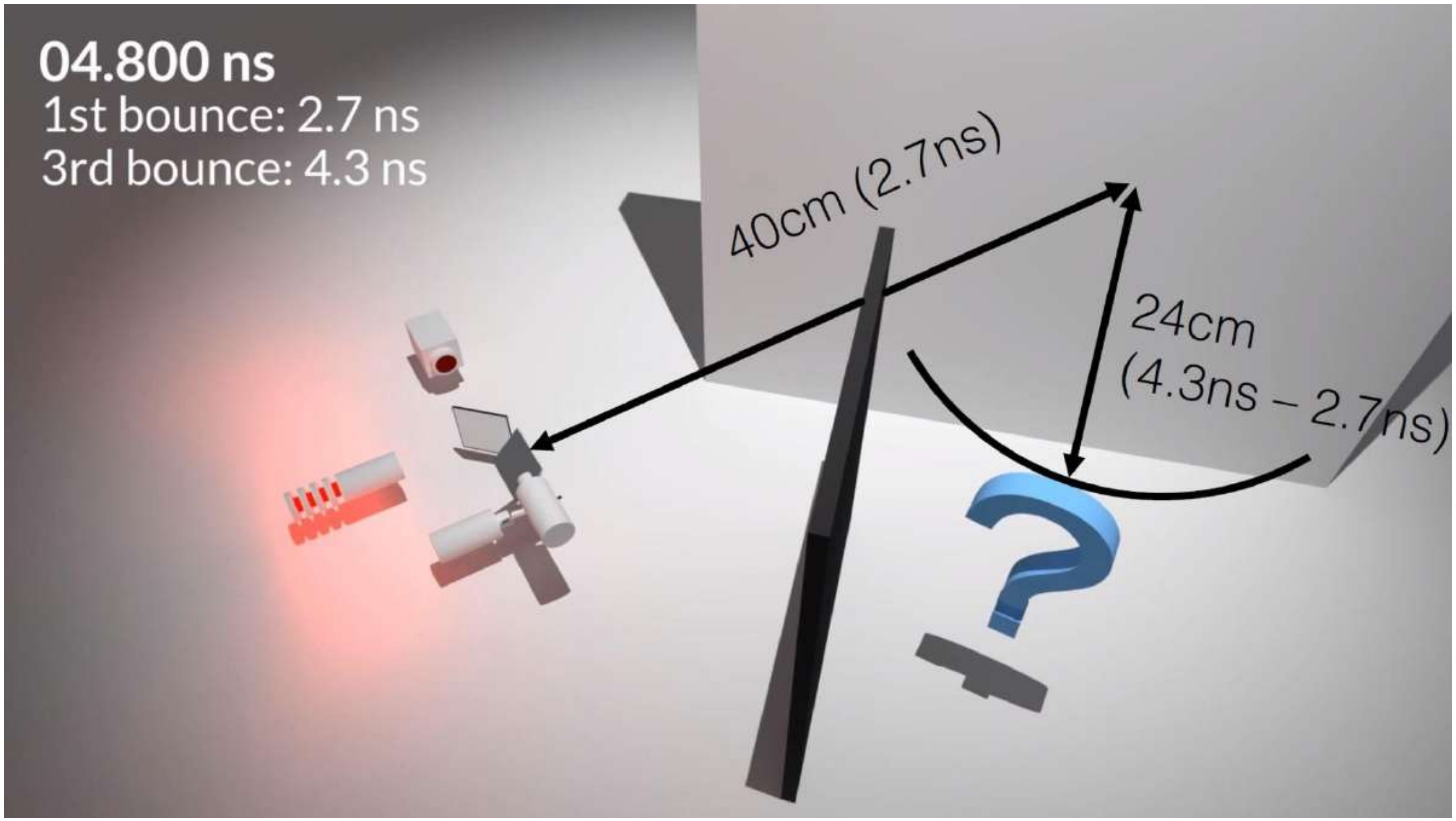

Motivating examples: non-trivial applications

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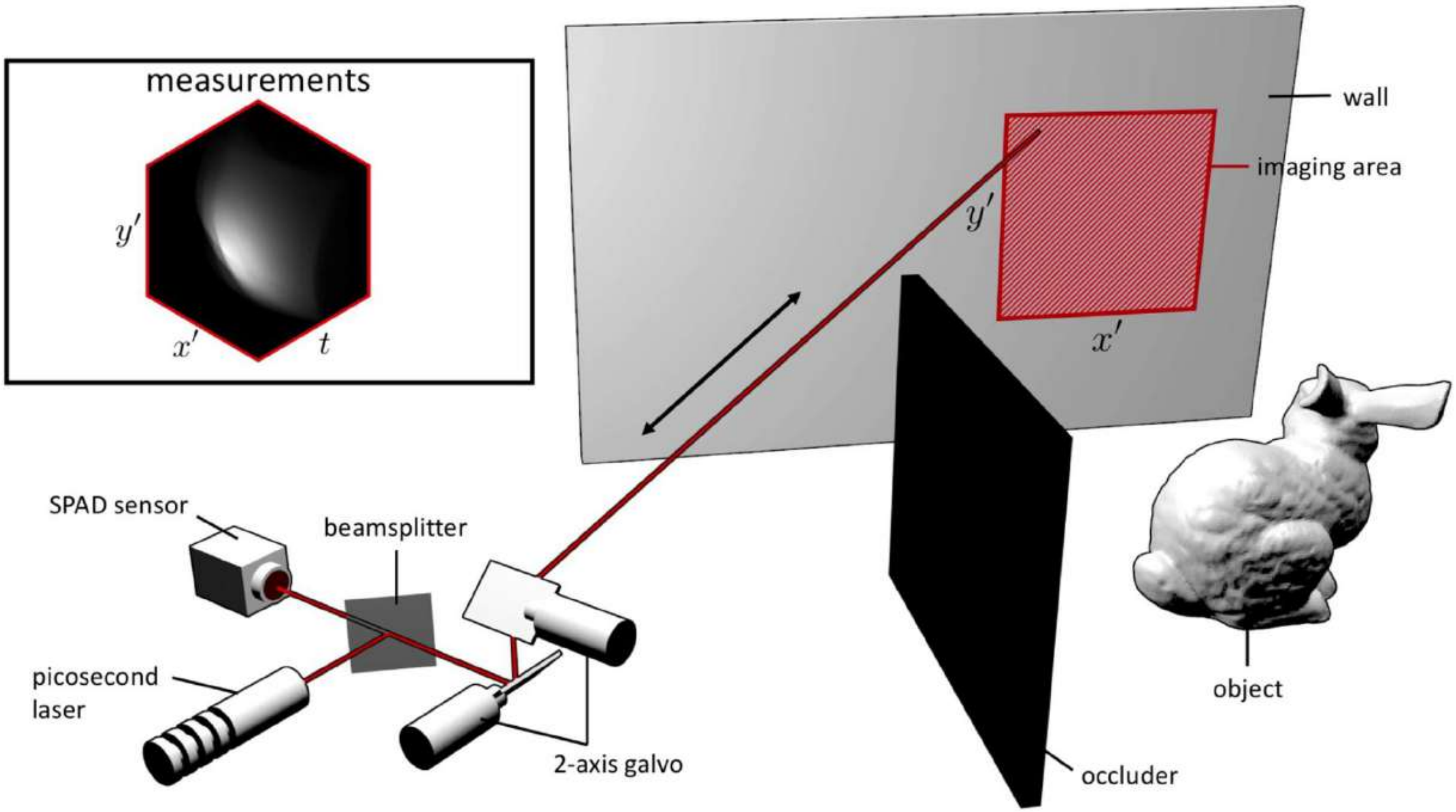

Motivating examples: non-trivial applications

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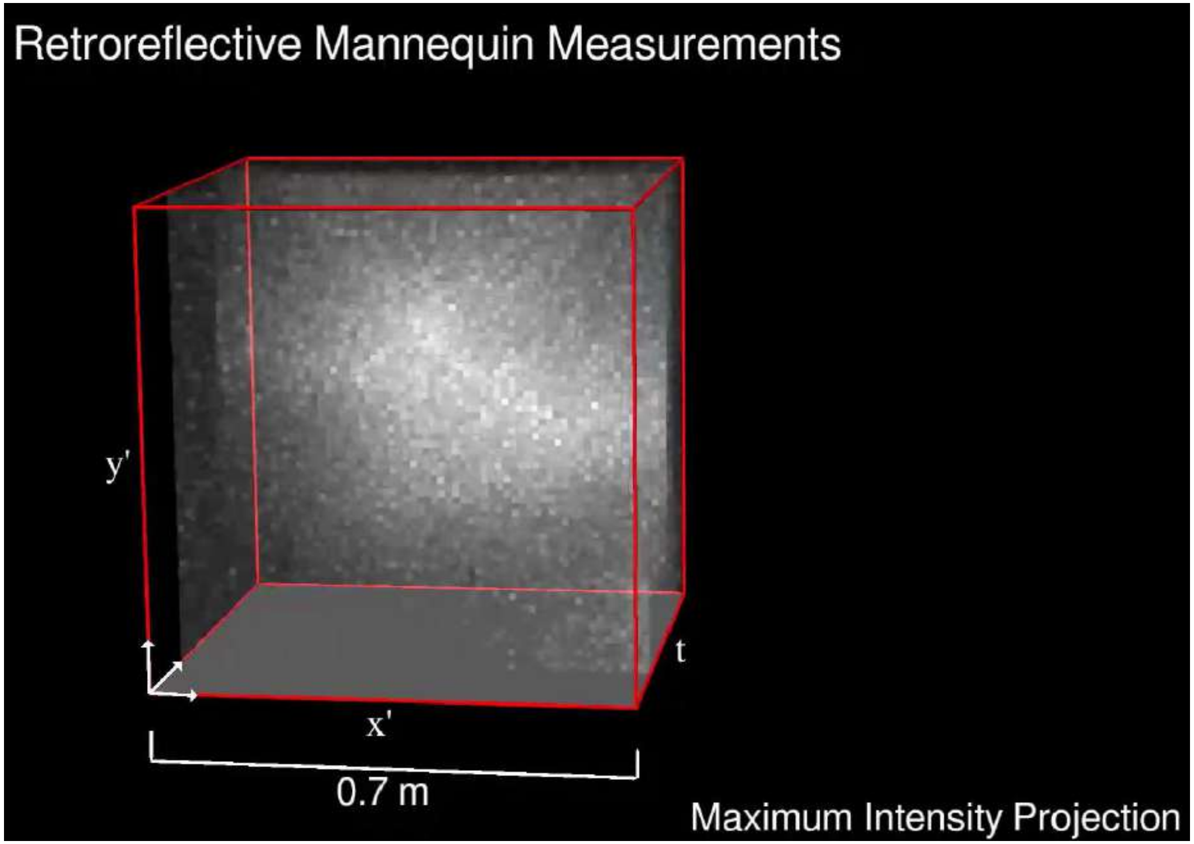

Motivating examples: non-trivial applications

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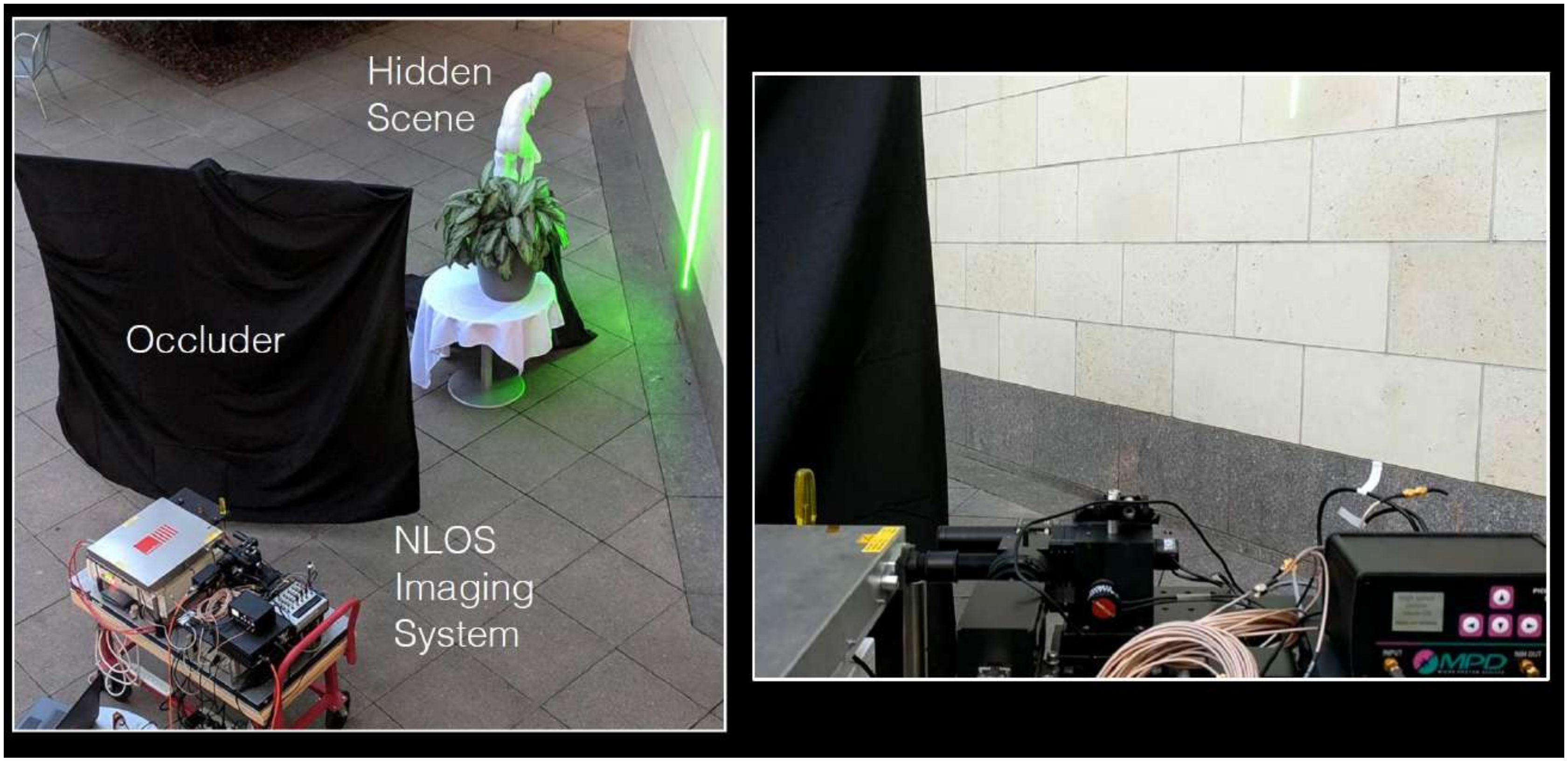
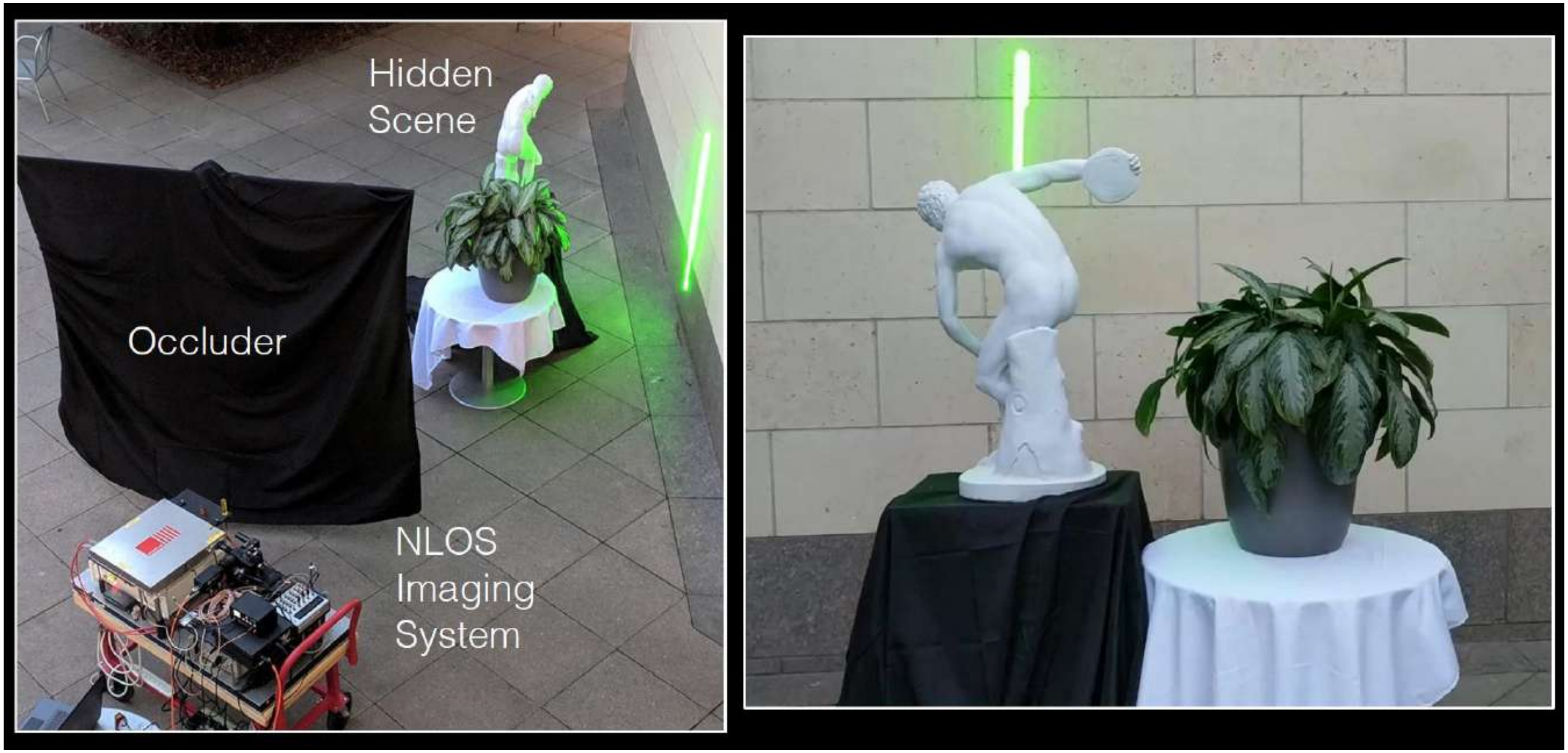


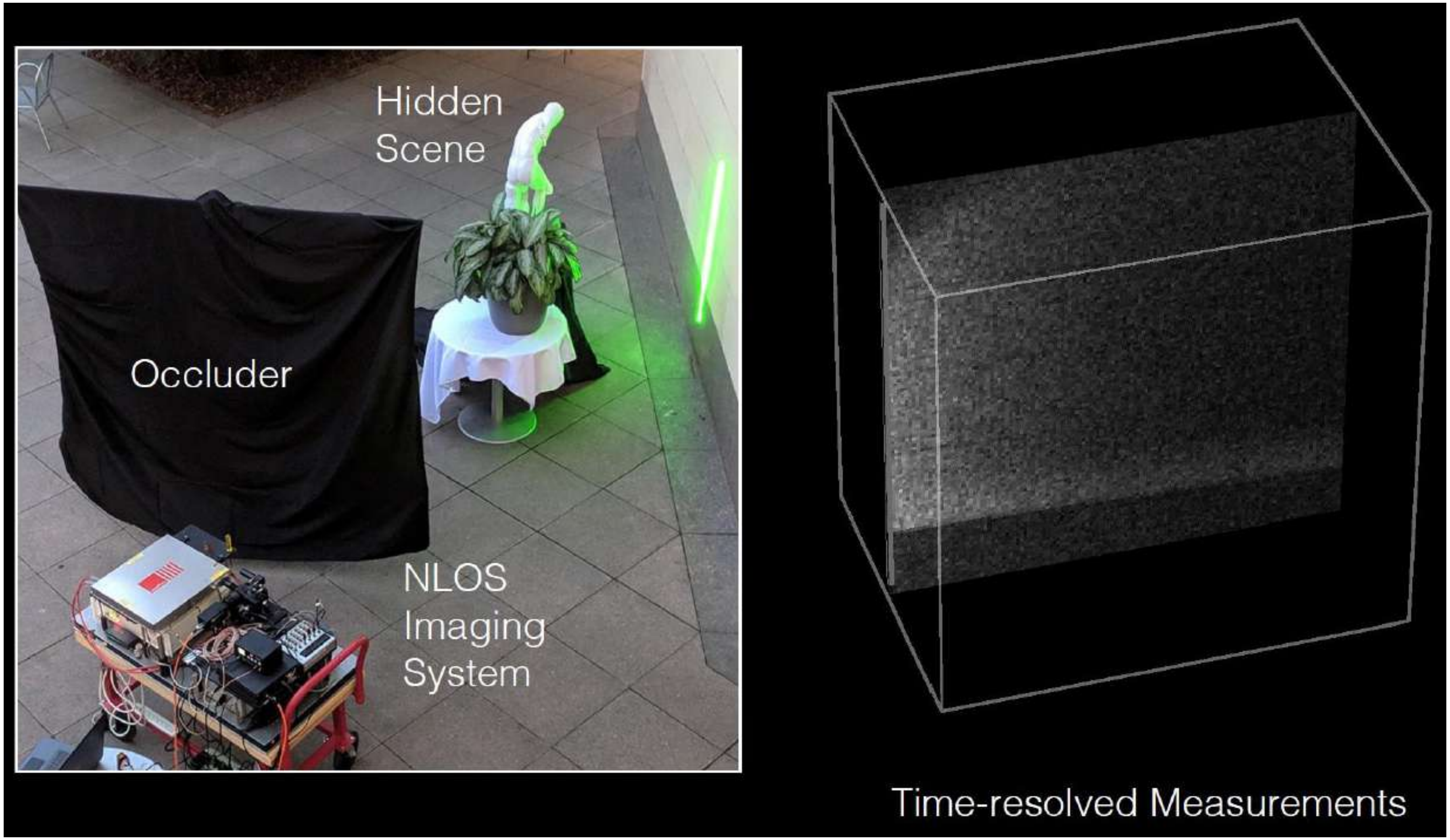
Motivating examples: non-trivial applications

Non-line-of-sight (NLOS) imaging



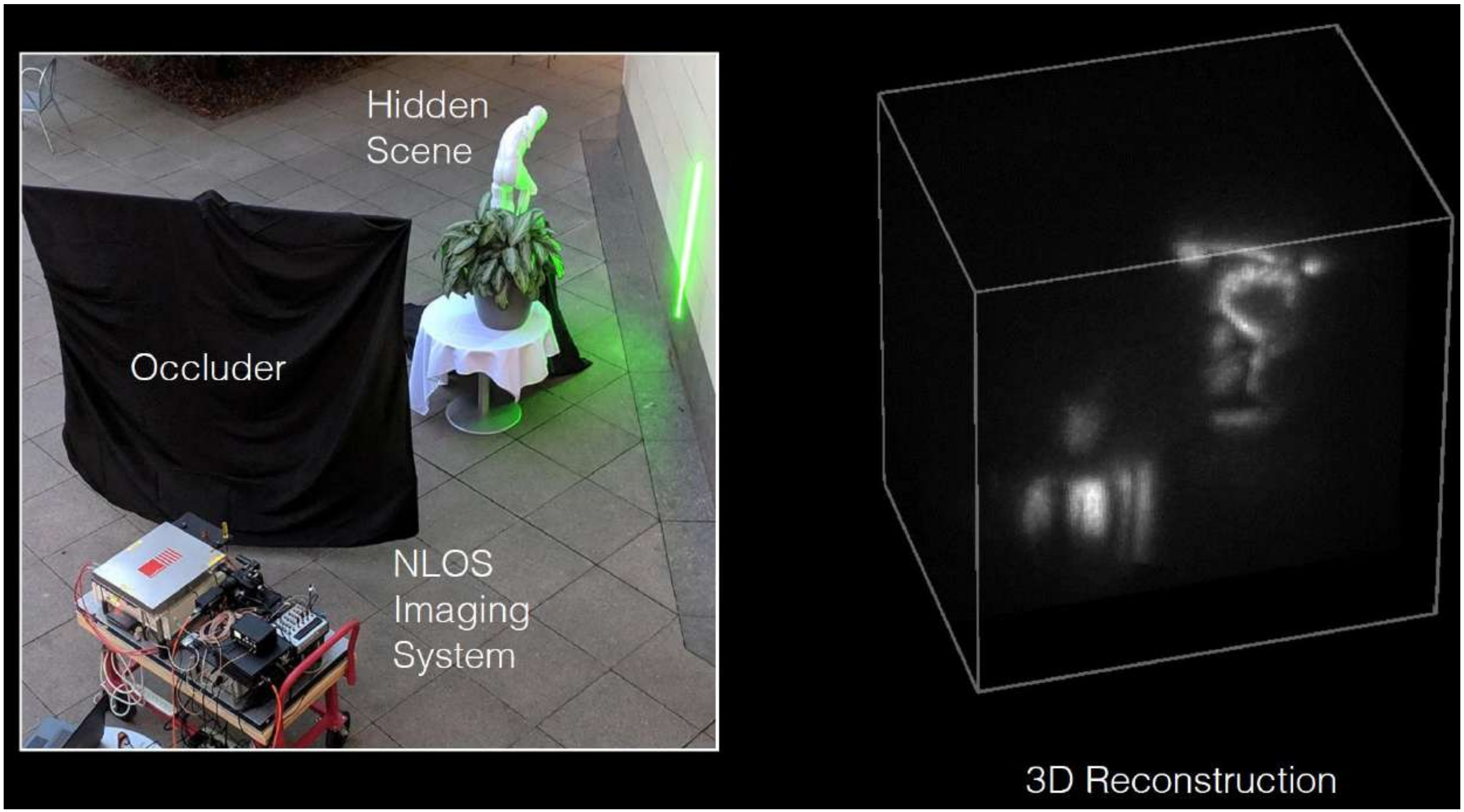
Motivating examples: non-trivial applications

Non-line-of-sight (NLOS) imaging



Motivating examples: non-trivial applications

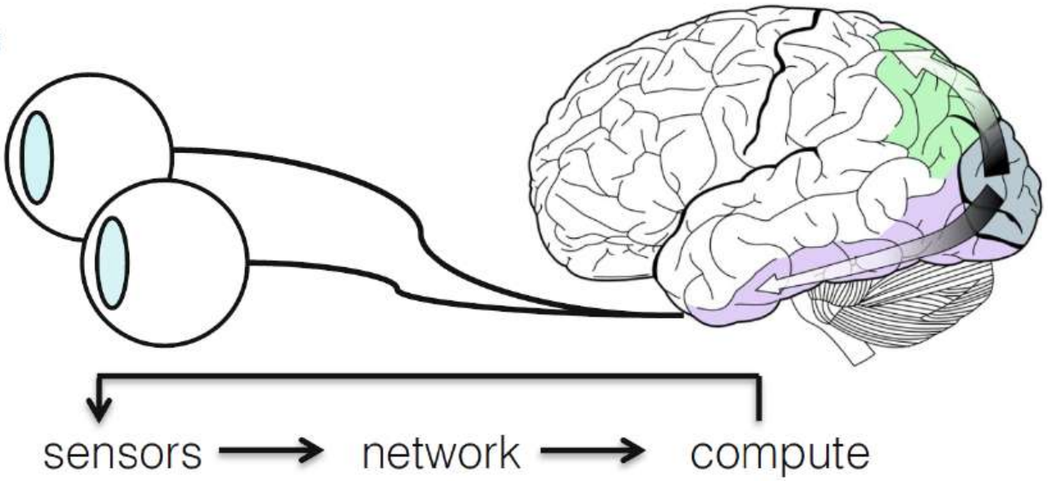
Non-line-of-sight (NLOS) imaging



Main topics of computational imaging

Human visual system

- anatomy of the eye
- acuity, color, 3D vision
- contrast sensitivity
- conflicts in displays
- refractive errors



Main topics of computational imaging

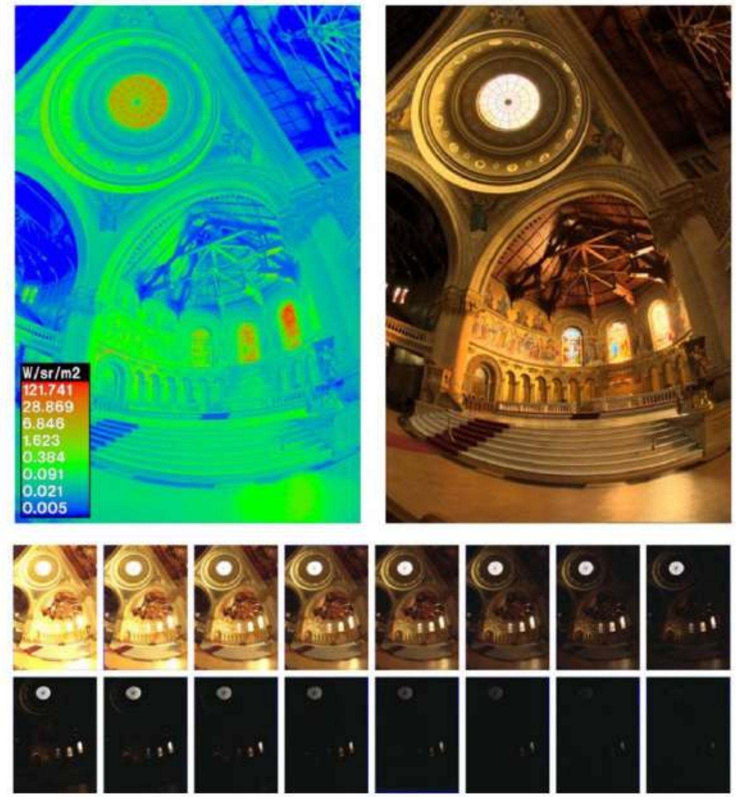
Digital photography

- optics
- aperture
- depth of field
- field of view
- exposure
- noise
- color filter arrays
- imaging processing pipeline

Main topics of computational imaging

Computational photography

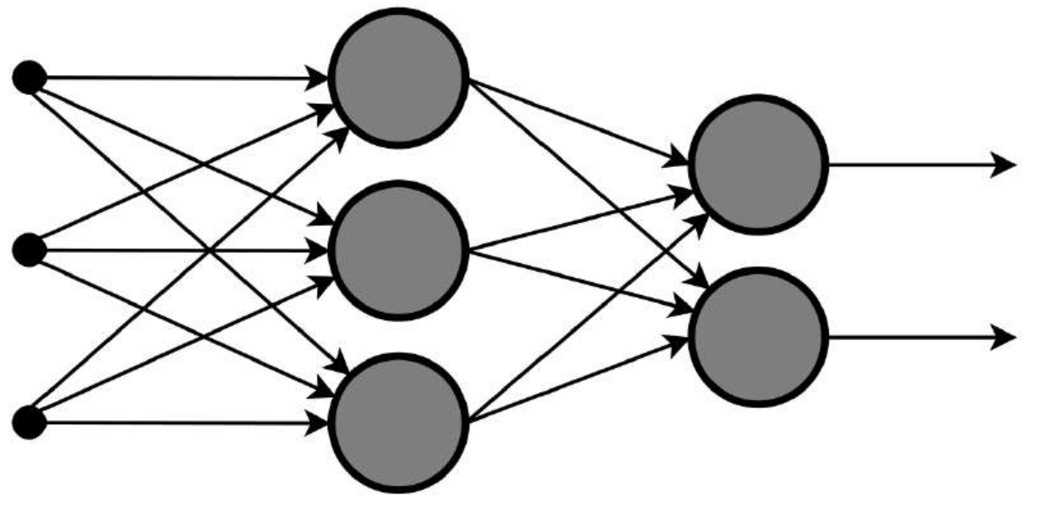
- High-dynamic range imaging
- Tone mapping
- Burst photography & night sight
- Coded apertures
- ...



Main topics of computational imaging

Deep learning for computational imaging

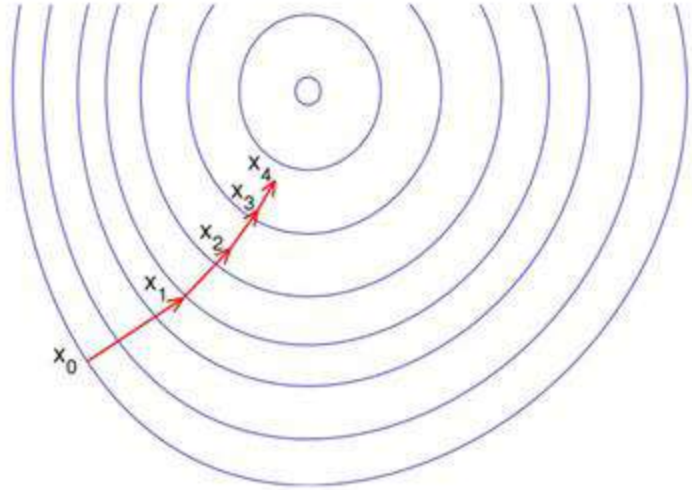
- Convolutional neural networks
- DnCNN
- U-Net
- …



Main topics of computational imaging

Optimization and deep learning

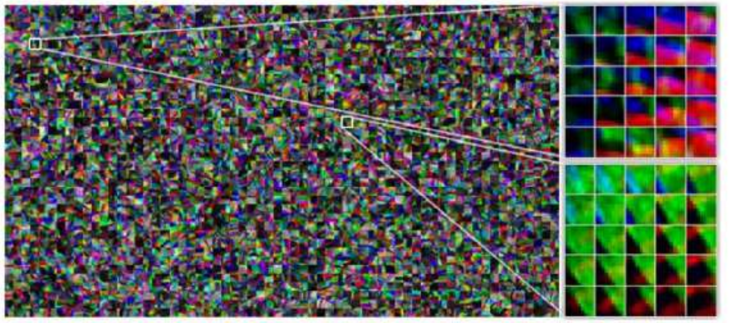
- Proximal gradient methods
- Iterative optimization with deep priors
- Solving general inverse problems in imaging
- …

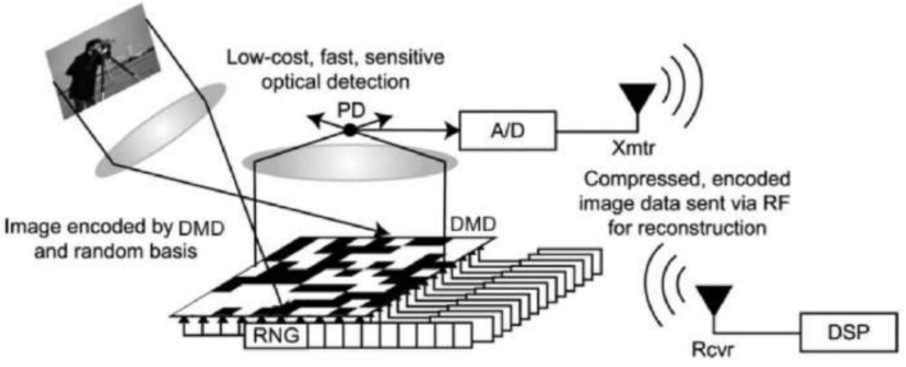


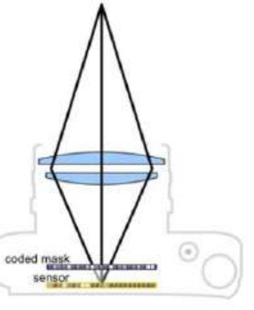
Main topics of computational imaging

Compressive imaging

- single pixel camera
- compressive hyperspectral imaging
- compressive light field imaging
- ...



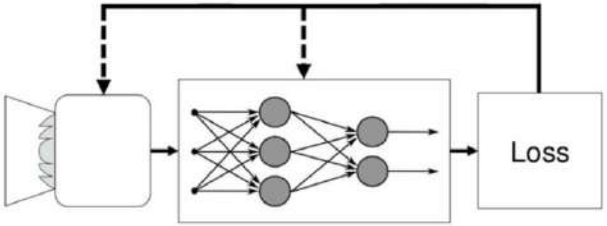


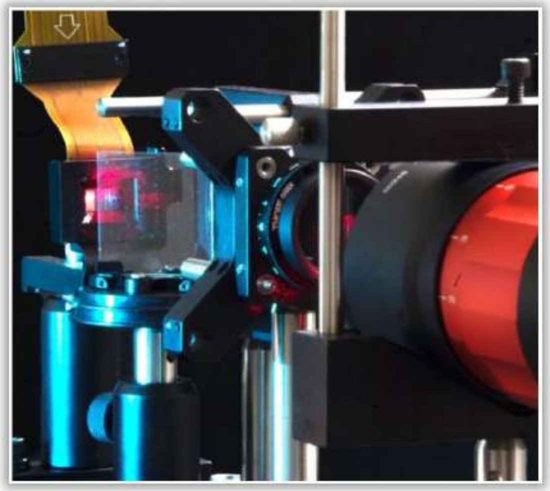


Main topics of computational imaging

Introduction to wave optics and deep optics

- Diffraction & interference
- The diffraction limit
- End-to-end optimization of optics & image processing
- Phase retrieval
- Computer-generated holography





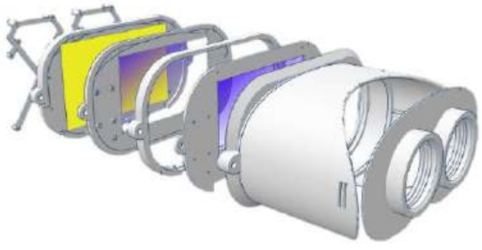
Main topics of computational imaging

Computational displays

Gaze-contingent Focus Displays

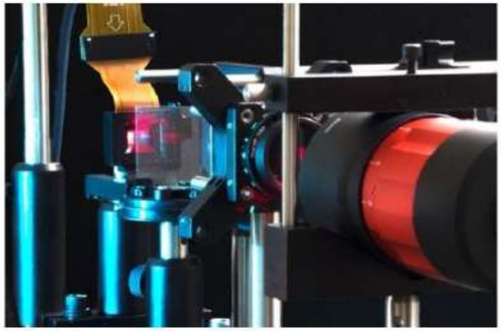
Konrad et al., SIGCHI 2016;
Padmanaban et al., PNAS 2017

Near-eye Light Field Displays



Huang et al., SIGGRAPH 2015;
Wetzstein et al., SIGGRAPH 2011, 2012

Holographic Displays



Padmanaban et al., SIGGRAPH Asia 2019
Peng et al., SIGGRAPH Asia 2020
Choi et al., Optica 2021

References

- Computational Photography (Levoy, Adams, Pulli, Stanford)
- Computational Photography SIGGRAPH Course (Raskar & Tumblin)
- Digital and Computational Photography (Durand & Freeman, MIT)
- Computational Photography (Efros, CMU)
- Computational Photography (Gkioulekas, CMU)
- Computational Imaging (Wetzstein, Stanford)
- Computational Photography (Fergus, NYU)
- Computational Imaging (Dragotti, Imperial College)
- Computer Vision (Seitz & Szeliski, UWashington)
- Introduction to Visual Computing and Visual Modeling (Kutulakos, UToronto)